Account Intelligence SaaS

Detailed Task-wise Roadmap, Task 1 to Task 100

Built from the PRD, the multi-client service design, Architecture Specification v2.0 and the generic solo-project roadmap

Prepared	1 October 2026
Status	Draft for engineering and product-owner review
Scope	Platform that turns one company name into a graded, sourced, refreshable account brief for many clients, with an optional private-context vault
Method	Dependency-first. Deterministic core (tenancy, ledgers, grading, state machines, gate) is built before the model layer.
Evidence rule	Every status below is based only on what the four uploaded documents show. Nothing is marked DONE unless its exit criteria are visibly met.

Source documents audited

- Account Intelligence: Business use case, product requirements and how the current system works (PRD, 6 Sep 2026)
- Account Intelligence as a Service: Multi-client SaaS design, scheduled refresh and alerts, enhanced version (6 Sep 2026)
- Account Intelligence SaaS Architecture and Implementation Specification v2.0 (1 Oct 2026, draft)
- Solo Project Task-Wise Implementation Roadmap v1.0 (generic framework, no product detail)

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1. Executive summary and how to read this roadmap

The documents describe a capability that already works by hand (a Claude skill that produced nine graded account briefs in September 2026) and a plan to turn it into a multi-client service with scheduled refresh, alerts, billing and, later, a private-context vault. The proven part is the research process. The unproven part is everything around it: tenant isolation, durable execution, metering, scheduling and the client application.

This roadmap breaks that work into exactly 100 tasks. Every task has an objective, architecture intent, implementation, testing and failure cases, exit criteria, dependencies, a status that is checked against the documents, and a priority and effort. Dependencies only point to earlier tasks, so the order is executable top to bottom.

Headline findings

- Status today: 1 task DONE, 24 PARTIAL, 75 NOT STARTED. The only finished item is documentation of the current system; the platform itself is not built. The v2.0 starter code is compile-checked but was never run against live services.
- The Architecture Specification v2.0 has no deliverable builder, coverage tracker with proof, people list, job-posting classifier or gate service. These are the heart of the proven process (PRD sections 3.4 and 4.1), so this roadmap adds them as explicit tasks.
- The documents disagree on several decisions (grading labels, agent runtime, hosting, regions, team size, when scheduling arrives). Phase 0 resolves each one before any build work starts.
- Broker spend is 64-70% of modelled cost, so Phase 0 includes a broker-economics spike on 20 real accounts before more than a few weeks of build.
- Estimated effort is about 411 person-days in total. That is well above the service design's own estimate of 8-12 weeks with one full-stack engineer for Phase 1 (an assumption in that document). See section 5 for scenarios.

Reading a task card

Status	DONE: all exit criteria are visibly met. PARTIAL: code, SQL or a reusable asset exists for part of the task, but exit criteria are not met. NOT STARTED: at most a prose design or a manual reference process exists.
Priority	P0: on the critical path or blocks a phase exit. P1: needed for the phase exit. P2: important, can slip one phase. P3: optional or conditional.
Effort	S: 1 day or less. M: 2-3 days. L: 4-7 days. XL: over 7 days. For totals the roadmap uses 1, 3, 6 and 10 days. These are planning estimates for one engineer, not commitments.
Phase	Phase 0 decide and prove; Phase 1 concierge service; Phase 2 self-serve; Phase 3 enhanced version with private material (matches the phases in both design documents).
Dependencies	Task numbers that must be DONE before the task starts. Tasks with no listed dependency can start immediately.

2. Audit of the source documents

What each document contributes

Document	Contributes	Limits
PRD and current system	The business problem, personas, 33 coverage rows, grading scheme (Confirmed, Probable, Hypothesis, Lead), seven deliverables, quality gate, standing rules, broker routing, refresh method, cost habit (one cap per run), known limitations.	Describes one operator working through claude.ai; no tenant concept; value figures are labelled assumptions.
Service design and enhanced version	Tenant model and isolation rules, component choices, run flow with parallel lanes, model-split question, refresh cadences, alert rules, credits and pricing, internal-document vault, two-output rule, three-phase roadmap, risks.	Effort estimates are assumptions; recommends Temporal or Inngest and several vendors without final choices.
Architecture Spec v2.0	Six isolation layers, RLS rules, durable slice execution, tool and wallet ledgers, grading maths, vector design, API catalogue, AI layer, infrastructure, CI/CD, SLOs, security, test and chaos plans, cost model, Phase 0 spikes, starter code.	Draft for review; 11 VERIFY items; assumes a six-engineer team and 28 weeks; code not executed; omits the PRD's deliverable pipeline.
Generic solo roadmap	A phase and gate structure (initiation to closure), risk and test templates, weekly reporting and handover checklists.	Contains no product requirements; useful only as a skeleton.

Audit summary

Built

- Research skill v1.5 with living checklist, lessons log, signals reference, tools playbook and output spec; nine accounts delivered.
- Gate scripts (rowcheck, doc_checks, third_party_scan, strip_tool_names), Word and Excel builder library, packaging into a workspace zip.
- Written design: PRD, service design, Architecture Spec v2.0 with ADRs, cost model, risk register and Phase 0 spike plan.

Partial

- Starter code in the spec (tenant_txn, RLS migration, ledgered_call, runner, Inngest loop, auth, run creation, isolation tests, CI, Dockerfile, compose). Compile-checked, never executed.
- Logical data model exists but lacks tables for coverage rows, people, job postings, technology detections, deliverables and tenant profile.

Not started

- Every platform component: infrastructure, database, tenancy tests, state machine, grading code, broker adapters, agents, API, portal, console, scheduler, alerts, billing, observability, vault.
- All five Phase 0 spikes, the golden set, the evaluation harness, and the verification register (V1-V11).

Risks

- Cross-tenant leakage; broker cost above price; broker resale terms; prompt injection; unproven workflow-engine limits; quality drop when moving from one model and one operator to parallel agents; overbuilding infrastructure for a small team.

Assumptions made in this roadmap

- Phase names and exit criteria follow the documents; where the two design documents differ, both criteria are kept.
- Effort is for one engineer per task; calendar time depends on team size (section 5).
- Model IDs named in the spec are illustrative and are re-verified in Phase 0 (V1).
- Legal, privacy and licence questions need counsel; this roadmap schedules the review but does not answer it.

Conflict register (input to the conflict-resolution task)

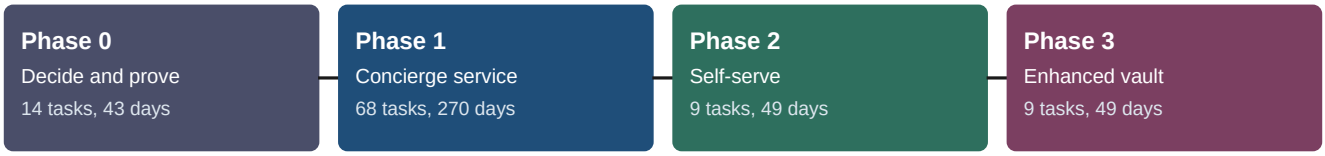
Topic	Document A	Document B	Roadmap handling
Grading labels	PRD: Confirmed, Probable, Hypothesis, Lead; only Confirmed may be said to a prospect	Spec: grades A-D plus Conflicting, Blocked, Unverified from a noisy-OR formula	Keep the numeric engine; map to the four client labels (taxonomy task)
Agent runtime	Service design: Claude Agent SDK with skill files mounted	Spec: LangGraph with Postgres checkpoints	Decided in the ADR task; spikes test the Spec's choice
Orchestrator	Service design: Temporal or Inngest	Spec: Inngest for trigger and slice loop; Temporal as alternative	Spike 2 decides; fallback is Temporal
Hosting	Service design: Vercel, Supabase or Neon, WorkOS or Clerk, S3 or R2	Spec: AWS EKS, Aurora, Karpenter, Argo CD, Mumbai plus Hyderabad	ADR picks the minimum viable setup; the Spec itself defers DR to V1

Topic	Document A	Document B	Roadmap handling
Regions	Service design: US and EU storage options	Spec: Mumbai primary, Hyderabad standby	Product-owner decision; vault task honours it
Team and time	Service design: 8-12 weeks, one full-stack engineer plus process owner (Phase 1)	Spec: six engineers, 28 weeks; generic roadmap: solo, 12 weeks	Section 5 computes scenarios from task effort
Scheduler phase	Service design: scheduler and weekly digest in Phase 1	Spec: scheduled probes and alerts in V1 (Phase 2)	Scheduler and alert engine v1 sit in Phase 1 to meet the pilot exit
Phase 0 spikes	Spec chapter 17: four spikes	Spec 19.2: five spikes (adds autoscaling)	Autoscaling burst test folded into the Kubernetes task
Deliverables	PRD: Word and Excel file set, zip, internal-only file, gate service	Spec: renders a brief; no file pipeline	Deliverables and gate tasks added

3. Roadmap overview

Phases, milestones and exit gates

Phase	Tasks	Count	Effort (days)	Exit milestone
Phase 0: Decide and Prove	1-14	14	43	All five spike pass criteria met, or the design is amended and the Phase 1 plan approved.
Phase 1: Concierge Service	15-82	68	270	Two paying tenants live, zero isolation defects, one month of on-time scheduled refresh; grade calibration measured; operators have run the parity set and 100 accounts overall.
Phase 2: Self-Serve	83-91	9	49	Runs start without an operator; ten paying tenants; SLOs met for 30 days; DR rehearsal passed.
Phase 3: Enhanced (Private Vault)	92-100	9	49	Leakage canaries at zero; retrieval recall targets met; one tenant in production on internal material with a completed security review; SOC 2 Type I issued or scheduled.
Total	1-100	100	411	



Effort by workstream

Workstream	Tasks	Days
Decisions & Spikes	14	43
Foundations	7	21
Tenancy & Data	8	23
Deterministic Core	12	40
Broker Layer	8	27
Evaluation	2	9
Agent Layer	12	57
Deliverables & Gate	5	21
API & Apps	3	18
Scheduler & Alerts	2	12
Release & Operations	7	36
Docs & Handover	3	9
Self-Serve	9	49
Private Vault	8	46

4. Schedule scenarios

Calendar time depends on how many engineers work in parallel and how much of each week is focused build time. The table below uses effort per task from this roadmap, a 75% focus factor, and the longest dependency chain inside each phase as the lower bound. These are planning estimates, not commitments.

Phase	Effort (days)	Longest chain (days)	1 engineer (weeks)	2 engineers (weeks)	6 engineers (weeks)
Phase 0: Decide and Prove	43	19	11	6	4
Phase 1: Concierge Service	270	128	72	36	26
Phase 2: Self-Serve	49	10	13	7	2
Phase 3: Enhanced (Private Vault)	49	49	13	10	10
Total	411	202	110	58	41

How this compares with the documents: the Architecture Spec plans 28 weeks with six engineers; the service design estimates 8-12 weeks for Phase 1 with one full-stack engineer plus a process owner; the generic roadmap assumes 12 weeks solo. The task-level effort here suggests the Phase 1 scope in the service design is under-estimated for a single engineer unless infrastructure, the deliverable pipeline or the evaluation work is cut or bought. Phase 0 spike results should be used to re-baseline these numbers.

Critical path (longest dependency chain by effort)

The chain below is computed from the task dependencies and effort; it totals 202 days. Any slip on these tasks moves the end date.

Order	Task	Title	Effort
1	T1	Business case, PRD and current-system documentation	S
2	T2	Service design and Architecture Specification v2.0 review	M
3	T3	Unified requirements baseline with REQ-IDs	M
4	T4	Source-document conflict register and resolution	S
5	T5	Product-owner decisions on pricing, scope and policy	S
6	T8	Lock ADRs and the minimum viable infrastructure	M
7	T10	Spike 1: tenant isolation with forced RLS and pooled connections	L
8	T14	Phase 0 gate review and design amendment	S
9	T15	Monorepo, tooling and local environment	S
10	T17	Infrastructure as code baseline	L
11	T18	Managed Postgres, roles and extensions	M
12	T22	Core schema and expand/contract migrations	L
13	T23	Forced RLS policies, fail-closed accessor and migration lint	M
14	T24	Tenant-scoped database access layer	S
15	T30	Run lifecycle state machine	M
16	T31	Credit wallet ledger: reserve, commit, refund	L
17	T32	Budget guard and circuit breakers	M
18	T52	Model router, tier configuration and fallback	M
19	T53	Prompt packaging, versioning and cache-aware layout	M
20	T54	Quarantine extractor for untrusted text	L
21	T55	Research graph skeleton	L
22	T56	Time-sliced runner, Inngest slice loop and reconciler	L
23	T58	Agents: set-up, relationship history and company baseline	L
24	T59	Agents: news, jobs and technology stack	L
25	T61	Verification node: Leads, citations and contradictions	M
26	T62	Distillation and writing-rule engine	L
27	T64	Document builder: Account Summary and Sources and Evidence	L
28	T65	Workbook builders: People List, Job Postings, Tech Stack and Teams	M

Order	Task	Title	Effort
29	T66	Workspace package and Internal-Only file split	M
30	T67	Gate service: pre-delivery checks and loop-back	L
31	T69	Core API v1	L
32	T74	Observability baseline	L
33	T76	Load and chaos test suite	L
34	T79	Production deployment with canary and rollback	L
35	T80	Technical documentation and runbooks	M
36	T82	Concierge pilot launch with two paying tenants	L
37	T87	Tenant data governance: export, deletion, offboarding and audit viewer	L
38	T92	Per-account vault, tenant keys and bridge and silo tiers	L
39	T93	Internal-source connectors	XL
40	T94	Ingestion pipeline: scan, parse and transcribe	L
41	T95	Redaction policy tiers and redaction maps	M
42	T96	Chunking, embeddings and hybrid retrieval	L
43	T97	Internal fact extraction and relationship state	M
44	T98	Two-output rule: Internal Only view and leakage scan	L
45	T99	Enhanced-version security review and production pilot	L
46	T100	Handover, maintenance plan and 30/60/90-day plan	M

5. Task catalogue

Each card follows the same template: objective, architecture intent, implementation, testing and failure cases, exit criteria, dependencies, status with evidence, priority and effort.

Phase 0: Decide and Prove

Exit milestone: All five spike pass criteria met, or the design is amended and the Phase 1 plan approved.

Workstream: Decisions & Spikes

Task 1: Business case, PRD and current-system documentation

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort S

OBJECTIVE	Record the proven manual capability as the reference behaviour the platform must reproduce.
WHY / ARCHITECTURE INTENT	Without a written baseline there is no parity target; the rebuild could silently drop rules learnt over nine runs (coverage with proof, grading, gate checks).
IMPLEMENTATION	Reference set: skill abm-account-research v1.5 (SKILL.md, living checklist, lessons log, signals reference, tools playbook, output spec), four gate scripts (rowcheck, doc_checks, third_party_scan, strip_tool_names), the Word and Excel builder library, the 33-row coverage tracker (5 Pass 1 rows + 28 Pass 2 rows) and the four-level grading scheme (Confirmed, Probable, Hypothesis, Lead).
TESTING / FAILURE CASES	Re-read each PRD requirement against the skill files; confirm each has an observable behaviour in at least one delivered account file set.
EXIT / DEFINITION OF DONE	PRD published; coverage rows and grading scheme listed; nine delivered file sets archived in the ABM Research Drive folder. Met.
DEPENDENCIES	None
STATUS	**DONE**
STATUS EVIDENCE	PRD (6 Sep 2026) documents the use case, requirements, the skill-based system and nine account runs; the service-design document lists 'Phase 0, now: nine accounts delivered, checklist stable'. Documentation is complete. Still open inside it: the seller-usefulness metric is 'to be instrumented'.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 2: Service design and Architecture Specification v2.0 review

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Hold one reviewed architecture that engineering can build against.
WHY / ARCHITECTURE INTENT	The two design documents propose different stacks and phase scopes. Building before review risks rework in tenancy and durability, the two things that are hardest to retrofit.
IMPLEMENTATION	Review Spec v2.0 against the service-design document: components, six isolation layers, run lifecycle, grading model, data model, cost model. Record every ADR as Accepted, Changed or Deferred in a decision log. Issue v2.1 with edits only where a decision changed.
TESTING / FAILURE CASES	Walk the nine attack scenarios (A1-A9) and the ten risk-register rows through the design on paper; each needs a named control and an owning task in this roadmap.
EXIT / DEFINITION OF DONE	Spec v2.1 tagged; every ADR has a status and reason; review notes stored; every attack scenario and risk maps to at least one task.
DEPENDENCIES	Task 1 (Business case, PRD and current-system documentation)
STATUS	**PARTIAL**
STATUS EVIDENCE	Companion SaaS design (tenancy, components, 3-phase roadmap) and Architecture Spec v2.0 (19 defects fixed vs v1, ADR-001..010, starter code) exist. Spec status is 'Draft for engineering review'; 11 VERIFY items and 5 product-owner questions are open. Missing: review sign-off and a merged v2.1.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 3: Unified requirements baseline with REQ-IDs

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Produce one numbered requirement list covering both the proven process (PRD) and the platform (Spec).
WHY / ARCHITECTURE INTENT	PRD requirements (coverage with proof, grading, gate checks, standing rules) do not appear in the v2 functional list. Untraced requirements are the ones dropped in a rebuild.
IMPLEMENTATION	Assign REQ-IDs to every PRD bullet and keep FR-01..FR-16 as aliases. Columns: acceptance condition, owning task, test ID. Keep the 33 coverage rows as tenant-configurable data, not as requirements. Store as docs/requirements.csv in the repo with a checker script.
TESTING / FAILURE CASES	Checker fails CI if a requirement lacks an owning task or acceptance condition. Hand-check ten PRD bullets for fidelity.
EXIT / DEFINITION OF DONE	Zero orphan requirements; each in-scope requirement has an acceptance condition and a reserved test ID; changes need a change-request entry.
DEPENDENCIES	Task 1 (Business case, PRD and current-system documentation), Task 2 (Service design and Architecture Specification v2.0 review)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec v2.0 lists FR-01..FR-16 and a measurable NFR table; the PRD has about 40 untagged requirement bullets (sections 3.3 to 3.5) and no traceability matrix. Missing: REQ-ID scheme, merged list, mapping to tasks and tests.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 4: Source-document conflict register and resolution

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort S

OBJECTIVE	Resolve every contradiction between the source documents with a dated decision.
WHY / ARCHITECTURE INTENT	Unresolved conflicts surface as mid-build rework, for example a grading model built twice or a hosting stack chosen by default.
IMPLEMENTATION	One row per conflict: options, evidence, decision, decider, date, tasks affected. Cover: grading labels (Confirmed..Lead vs A-D), agent runtime (Claude Agent SDK vs LangGraph), orchestrator (Temporal or Inngest), hosting (Vercel/Supabase/Neon vs AWS EKS/Aurora), residency (US/EU vs Mumbai/Hyderabad), team and duration, scheduler phase, and the missing deliverable builder in Spec v2.0.
TESTING / FAILURE CASES	Every row has a decision or an explicit deferral with a trigger; no later task references an undecided conflict.
EXIT / DEFINITION OF DONE	All conflict rows closed or deferred with trigger; register linked from the ADR log.
DEPENDENCIES	Task 3 (Unified requirements baseline with REQ-IDs)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The four documents disagree on grading labels, agent runtime, orchestrator, hosting, regions, team size and scheduler phase (see the Audit section). No resolution is recorded anywhere.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 5: Product-owner decisions on pricing, scope and policy

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort S

OBJECTIVE	Answer the business questions that gate design and pricing.
WHY / ARCHITECTURE INTENT	Credit unit, depth tiers, residency and people-data scope change the data model, billing and compliance work.
IMPLEMENTATION	Decision record covering: list price per run or credit and depth tiers; regions for the first ten customers and whether any need a silo tier on day one; people-data attributes allowed in briefs; operator console permanent or transitional; first Phase B connectors; written position on two tenants researching the same account; whether 6th tells anyone.
TESTING / FAILURE CASES	Each answer is checked against the break-even table (Spec 16.2) and the privacy review; contradictions are escalated.
EXIT / DEFINITION OF DONE	Written decisions signed by the product owner; price per run for 70% gross margin recomputed.
DEPENDENCIES	Task 4 (Source-document conflict register and resolution)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 19.3 lists five open questions and the service design lists more (BitScale vs Sumble, two tenants on one account, model split). None is answered in the documents.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 6: Commercial and legal clearance (broker terms, privacy law, DPA)

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Confirm the business may lawfully collect, derive and resell what the design promises.
WHY / ARCHITECTURE INTENT	Brokers are 64-70% of cost, and the product handles business contact data about identifiable people. A licence or privacy ruling after the build can invalidate the model.
IMPLEMENTATION	Review terms of Parallel, Firecrawl, PredictLeads, BitScale and Sumble for redistribution of derived data; review website terms for scraped sources; lawful-basis analysis, notice and objection handling for people data under India's DPDP Act and, where applicable, GDPR; draft DPA and sub-processor register (the only place brokers are named to a client); confirm zero-retention and no-training terms for LLM, embedding and speech vendors. This is not legal advice; counsel signs off.
TESTING / FAILURE CASES	Map each clause to a design rule (what may be cached, shared, shown or exported); list clauses with no design answer.
EXIT / DEFINITION OF DONE	Counsel memo on file; DPA template and sub-processor register drafted; each design rule that depends on a term is tagged cleared or blocked.
DEPENDENCIES	Task 5 (Product-owner decisions on pricing, scope and policy)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Both design documents flag broker resale terms, DPDP/GDPR duties and vendor retention terms as unverified. No review is recorded.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 7: Close the verification register (V1-V11)

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Replace every [VERIFY] assumption with a first-party fact.
WHY / ARCHITECTURE INTENT	Model IDs, cache lifetimes, Aurora Global support, pgvector features, Inngest limits and checkpoint library behaviour each decide a design choice.
IMPLEMENTATION	For each item record the first-party source, date, owner and consequence: V1 model IDs, prices, cache TTL and multipliers; V2 Aurora Global in ap-south-2 and measured latency; V3 pgvector version and iterative scan; V4 Inngest limits and SDK signatures; V5 vendor retention terms; V6 speech pricing and diarization; V7 broker contracts; V8 Karpenter and spot savings; V9 OWASP edition; V10 DPDP/GDPR scope; V11 LangGraph checkpoint setup() and adelete_thread.
TESTING / FAILURE CASES	Pin versions in the repo and run a smoke script per item against those pins; any mismatch reopens the dependent ADR.
EXIT / DEFINITION OF DONE	All 11 closed, or accepted as a named risk with an owner; [VERIFY] tags removed from Spec v2.1.
DEPENDENCIES	Task 2 (Service design and Architecture Specification v2.0 review)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec Appendix B lists V1-V11 with blank owner and date columns.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 8: Lock ADRs and the minimum viable infrastructure

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Finalise the decisions that every later task builds on.
WHY / ARCHITECTURE INTENT	A solo or small team cannot carry six isolation layers, EKS, multi-region and a self-serve product at once; the minimum viable infrastructure must still keep forced RLS, per-run connections and KMS.
IMPLEMENTATION	Decide and record: orchestrator (Inngest + LangGraph checkpoint per Spec, Temporal as documented alternative); agent runtime; database host; identity provider (WorkOS, Clerk or another); vector store; portal hosting. Write a monthly cost estimate for the chosen minimum setup. Each ADR carries a 'revisit if' trigger tied to a spike result.
TESTING / FAILURE CASES	Each ADR is checked against the five spike pass criteria and the NFR table; reject any choice that weakens an isolation layer.
EXIT / DEFINITION OF DONE	ADR-001..010 final plus new ADRs for auth provider, hosting and agent runtime; cost estimate approved.
DEPENDENCIES	Task 4 (Source-document conflict register and resolution), Task 5 (Product-owner decisions on pricing, scope and policy), Task 7 (Close the verification register (V1-V11))
STATUS	**PARTIAL**
STATUS EVIDENCE	ADR-001..010 summary exists in Spec 3.5; Spec says MVP relies on multi-AZ plus tested backups and DR starts at V1. Not decided: orchestrator, agent runtime, database host, auth provider, portal hosting.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 9: Golden set v0 from the nine delivered accounts

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort M

OBJECTIVE	Create the first labelled evaluation set, so every later change has a baseline.
WHY / ARCHITECTURE INTENT	Real-world facts decay; a stale label punishes a correct agent. The set must carry capture dates and be stratified, and it should exist before the first graph.
IMPLEMENTATION	Select the nine accounts, strip relationship history and any internal material (public facts only), record capture dates, label expected attributes and grade-A/B facts from the delivered Evidence files, store broker responses as replay fixtures, define the labelling and quarterly re-verification protocol, stratify by size, industry and region and plan the extension to 50.
TESTING / FAILURE CASES	Two reviewers label a sample and agreement is measured; spot-check labels against live sources; confirm no private or client-identifying data is in fixtures.
EXIT / DEFINITION OF DONE	Nine accounts labelled with capture dates; fixtures load into the eval runner; labelling protocol written; agreement on the sample at or above 0.8.
DEPENDENCIES	Task 1 (Business case, PRD and current-system documentation)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Nine graded file sets exist (reference), but nothing is labelled for evaluation: no capture dates, no label protocol, no fixtures. Spec target is 50 accounts at MVP growing to 200.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 10: Spike 1: tenant isolation with forced RLS and pooled connections

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort L

OBJECTIVE	Prove tenant isolation holds under connection pooling before any feature work.
WHY / ARCHITECTURE INTENT	Isolation is the paramount constraint and the hardest to retrofit; a failure here changes tenancy defaults.
IMPLEMENTATION	Build forced RLS on a small schema with non-owner role, fail-closed accessor, transaction-local set_config, checkpoint prefix policy and hash-partitioned vectors with planted canary vectors; drive 100 concurrent mixed-tenant requests over a 5-connection pool.
TESTING / FAILURE CASES	Failure cases: SET LOCAL outside a transaction, transaction-mode pooler pinning, BYPASSRLS role, owner bypass, policy wrapper changing the plan, filtered ANN under-recall for small tenants.
EXIT / DEFINITION OF DONE	Zero cross-tenant rows across the run; unset context returns zero rows; canary vectors never returned to another tenant. If it fails, tenancy tier defaults are revisited before Phase 1.
DEPENDENCIES	Task 8 (Lock ADRs and the minimum viable infrastructure)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 19.2 defines the spike and pass criterion; starter code is compile-checked only and has not run against a live database.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 11: Spike 2: durable slices, resume and the tool ledger

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort L

OBJECTIVE	Prove a run survives repeated process kills without paying twice.
WHY / ARCHITECTURE INTENT	Multi-hour runs call paid APIs; duplicated spend or lost work breaks unit economics and trust.
IMPLEMENTATION	Two-node graph with a fake paid tool behind the ledger; time-sliced runner (240 s default) driven by a local Inngest dev server; kill the process at random points during fan-out; also pause a run longer than the prompt-cache lifetime.
TESTING / FAILURE CASES	Failure cases: crash between vendor call and ledger insert (bounded to one repeat), duplicate event delivery, checkpoint written but slice result lost, SDK call-shape mismatch.
EXIT / DEFINITION OF DONE	Run completes after 5 random kills; ledger shows each paid call exactly once; no corrupted checkpoint. If it fails, evaluate Temporal for the outer loop.
DEPENDENCIES	Task 8 (Lock ADRs and the minimum viable infrastructure)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 19.2 defines the spike; runner and orchestration code exist only as unexecuted listings.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 12: Spike 3: prompt-cache ratio on real prompts

Phase 0: Decide and Prove | Decisions & Spikes | Priority P1 | Effort M

OBJECTIVE	Measure whether the cache-aware prompt layout reaches the assumed 75% read ratio.
WHY / ARCHITECTURE INTENT	The LLM cost model assumes a 75% cache-read ratio; a silent miss adds about 45% to LLM cost at 0%.
IMPLEMENTATION	Run the baseline prompt 10 times in a loop with realistic tool definitions; stable-prefix layout, canonical JSON, sorted tools, explicit cache breakpoints; test the longer-lifetime option and a warm-up call at slice start; record cache-read, cache-write and uncached tokens per call.
TESTING / FAILURE CASES	Failure cases: timestamp or request id in the prefix, unsorted tools, prefix below the minimum cacheable length, pause longer than the lifetime.
EXIT / DEFINITION OF DONE	Cache-read ratio at least 70% from the second call; no invalidation across slices of 6 minutes; cost model updated with measured values.
DEPENDENCIES	Task 8 (Lock ADRs and the minimum viable infrastructure), Task 7 (Close the verification register (V1-V11))
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 19.2 defines the spike; V1 (cache lifetime, minimum length, breakpoint limit) is unverified.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 13: Spike 4: broker economics on 20 real accounts

Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort L

OBJECTIVE	Learn what a standard-depth run really costs across the five brokers.
WHY / ARCHITECTURE INTENT	Brokers are 64-70% of total cost in the model; if cost per run exceeds the cap, pricing or scope must change before building.
IMPLEMENTATION	Run 20 accounts at standard depth through the five brokers behind a metering shim, using the nine delivered accounts as part of the sample; record cost per source group; compare BitScale-only vs BitScale plus Sumble for people; compute coverage of expected attributes and price per run for 70% gross margin.
TESTING / FAILURE CASES	Failure cases: out-of-credit broker mid-run, rate limits, expensive modes (JSON extraction 5 credits, search-with-scrape 20+), pre-pulled file fallback.
EXIT / DEFINITION OF DONE	Cost per run within the planned cap; coverage of expected attributes at least 0.80; people-source decision recorded with fill rate and cost per usable senior person.
DEPENDENCIES	Task 8 (Lock ADRs and the minimum viable infrastructure), Task 6 (Commercial and legal clearance (broker terms, privacy law, DPA)), Task 9 (Golden set v0 from the nine delivered accounts)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. PRD says broker spend has run at 'a few hundred credits across tools' per account (observed, varies); Spec assumes $4.50 per run at MVP. No measured cost-per-run exists.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 14: Phase 0 gate review and design amendment Phase 0: Decide and Prove | Decisions & Spikes | Priority P0 | Effort S	
OBJECTIVE	Make a go, amend or stop decision for Phase 1 using spike evidence.
WHY / ARCHITECTURE INTENT	Spec 17 says Phase 0 ends when all spike criteria are met or the design is amended; skipping this commits months to unproven assumptions.
IMPLEMENTATION	Review each spike against its pass criterion, record failures with the fallback from Spec 19.2 (revisit tenancy defaults, evaluate Temporal, adopt longer cache lifetime, reprice or narrow brokers), update the cost model and risk register, confirm team size and budget for Phase 1.
TESTING / FAILURE CASES	Gate record must cite measured numbers for every criterion; no criterion may be waived without a named owner.
EXIT / DEFINITION OF DONE	Signed gate record; Spec v2.2 amended where needed; Phase 1 plan and budget approved.
DEPENDENCIES	Task 10 (Spike 1: tenant isolation with forced RLS and pooled connections), Task 11 (Spike 2: durable slices, resume and the tool ledger), Task 12 (Spike 3: prompt-cache ratio on real prompts), Task 13 (Spike 4: broker economics on 20 real accounts)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Depends on spike results that do not exist yet.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Phase 1: Concierge Service

Exit milestone: Two paying tenants live, zero isolation defects, one month of on-time scheduled refresh; grade calibration measured; operators have run the parity set and 100 accounts overall.

Workstream: Foundations

Task 15: Monorepo, tooling and local environment Phase 1: Concierge Service | Foundations | Priority P0 | Effort S	
OBJECTIVE	Make a clean clone build, test and run in minutes.
WHY / ARCHITECTURE INTENT	Reproducible setup is the base for every later task and for CI.
IMPLEMENTATION	Create the layout (.github, deploy/terraform, deploy/helm, migrations, src/api|core|db|orchestrator|agents|ingestion|render, tests/isolation|unit|integration|contract|e2e|chaos|eval); uv with lockfile; ruff and mypy; pre-commit with secret scan; compose.yml with pgvector pg16, Redis 7 and the Inngest dev server; .env.example; CODEOWNERS on migrations.
TESTING / FAILURE CASES	Fresh-machine setup run by a second person; compose health checks; lockfile drift check.
EXIT / DEFINITION OF DONE	Fresh clone to green unit tests in under 30 minutes using documented steps; main branch protected.
DEPENDENCIES	Task 14 (Phase 0 gate review and design amendment)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.1 gives the repository layout and 18.10 gives Dockerfile, compose.yml and pyproject conventions. Nothing is created; the listings are unexecuted.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 16: CI pipeline v1 with hard isolation gate

Phase 1: Concierge Service | Foundations | Priority P0 | Effort S

OBJECTIVE	Block any change that breaks isolation, types or tests.
WHY / ARCHITECTURE INTENT	The isolation suite is a hard gate; a pipeline that does not run it makes tenancy a convention.
IMPLEMENTATION	GitHub Actions: lint, types, migrate a scratch database, run isolation tests first with -x, unit tests with 80% coverage and 100% on grading and billing, image build, Trivy HIGH/CRITICAL failure, secret scan; actions pinned by commit SHA; eval-gate job on main.
TESTING / FAILURE CASES	Deliberately break an RLS policy and a type annotation; confirm the pipeline fails for each.
EXIT / DEFINITION OF DONE	Required checks enforced on main; median pipeline time under 10 minutes; broken-policy canary fails the build.
DEPENDENCIES	Task 15 (Monorepo, tooling and local environment)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.10 contains a ci.yml draft (ruff, mypy, alembic upgrade, isolation tests, coverage 80%, docker build, Trivy). Not run.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 17: Infrastructure as code baseline

Phase 1: Concierge Service | Foundations | Priority P0 | Effort L

OBJECTIVE	Create dev and staging environments from code.
WHY / ARCHITECTURE INTENT	Manual infrastructure drifts and cannot be audited, which blocks SOC 2 and disaster recovery.
IMPLEMENTATION	AWS Organization with management, security and log archive, shared services, development, staging and production accounts; service control policies denying disabled logging and encryption; Terraform modules for network (private subnets, VPC endpoints for S3, ECR, STS, KMS, Secrets Manager), compute, database, S3 with KMS, IAM, observability; remote state with locking; plans in pull requests; nightly drift detection; region ap-south-1 unless the ADR changes it.
TESTING / FAILURE CASES	Destroy and recreate dev from scratch; policy scan with tfsec or checkov; test that an SCP blocks disabling CloudTrail.
EXIT / DEFINITION OF DONE	Dev and staging rebuild from one command; drift job green; zero high findings from the policy scan.
DEPENDENCIES	Task 8 (Lock ADRs and the minimum viable infrastructure), Task 15 (Monorepo, tooling and local environment)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 11.4 describes the AWS Organization layout and Terraform modules; no Terraform exists.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 18: Managed Postgres, roles and extensions

Phase 1: Concierge Service | Foundations | Priority P0 | Effort M

OBJECTIVE	Provide the database that enforces tenancy.
WHY / ARCHITECTURE INTENT	RLS is only as strong as the roles and connection handling around it.
IMPLEMENTATION	Aurora PostgreSQL multi-AZ with pgvector; roles: app_rw (LOGIN NOINHERIT NOBYPASSRLS), migrator (DDL only, never at runtime), public_reader (NOLOGIN); writer connection budget = (API pool x pods) + (worker concurrency x pods) + headroom; point-in-time recovery; startup self-check query that refuses to boot if the role bypasses RLS or owns tables.
TESTING / FAILURE CASES	Restore to a point in time into a scratch cluster; attempt to connect as app_rw and create a table; check owner bypass.
EXIT / DEFINITION OF DONE	Self-check passes in dev and staging; restore drill completed and timed; connection budget documented.
DEPENDENCIES	Task 17 (Infrastructure as code baseline)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 4.2 and 18.3 define roles app_rw, migrator and public_reader and a connection-sizing rule; nothing is provisioned.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 19: Secrets, workload identity and per-tenant key management

Phase 1: Concierge Service | Foundations | Priority P1 | Effort M

OBJECTIVE	Remove long-lived credentials and make tenant keys possible.
WHY / ARCHITECTURE INTENT	Customer OAuth tokens, broker keys and (later) private documents need key-level isolation and clean offboarding.
IMPLEMENTATION	Secrets Manager with External Secrets or CSI mounted as files; one narrow role per workload via Pod Identity or IRSA; KMS encryption context carrying the tenant id for S3 and token storage; automatic database credential rotation; broker keys rotated by calendar and on personnel change.
TESTING / FAILURE CASES	IAM policy simulation denies cross-prefix access; a pod without a role fails closed; canary secret strings never appear in logs.
EXIT / DEFINITION OF DONE	No long-lived cloud keys in the account; rotation job proven; tenant-context encryption demonstrated on a test object.
DEPENDENCIES	Task 17 (Infrastructure as code baseline)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 11.5 and 4.4 describe Secrets Manager, Pod Identity or IRSA, per-tenant KMS data keys and crypto-shredding.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 20: Kubernetes platform and autoscaling

Phase 1: Concierge Service | Foundations | Priority P1 | Effort L

OBJECTIVE	Run stateless workers that scale on queue age and survive spot reclamation.
WHY / ARCHITECTURE INTENT	Workers are I/O bound, so scaling on CPU is wrong; interruption must be safe by design.
IMPLEMENTATION	EKS with pools system (2 on-demand), api (on-demand, PodDisruptionBudget max 1), agents (spot first, 120 s grace), ingestion (spot), jobs (do-not-disrupt); Karpenter; Argo CD; default-deny NetworkPolicies; scaling on queue age and in-flight runs. Skip or replace with a managed platform if the ADR chooses one for Phase 1.
TESTING / FAILURE CASES	Burst of 50 simultaneous runs against stub brokers; terminate most agent nodes; SIGTERM during a slice.
EXIT / DEFINITION OF DONE	Pods scheduled within a minute; consolidation after the burst; no run lost on spot reclamation; measured spot saving recorded (not assumed).
DEPENDENCIES	Task 17 (Infrastructure as code baseline), Task 15 (Monorepo, tooling and local environment)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 11.1 defines node pools, Karpenter, disruption policy and graceful shutdown; spike 5 (autoscaling burst) appears only in Spec 19.2, not in the Phase 0 list in chapter 17.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 21: Container build, signing and supply chain

Phase 1: Concierge Service | Foundations | Priority P1 | Effort S

OBJECTIVE	Ship only signed, scanned, reproducible images.
WHY / ARCHITECTURE INTENT	Unsigned or unscanned images undermine every other control and SOC 2 evidence.
IMPLEMENTATION	Multi-stage build with uv frozen lockfile; pinned base digests; cosign signature; SBOM per release; Trivy scan; registry by digest; admission policy rejecting unsigned images; dependency update bot with reviewed PRs.
TESTING / FAILURE CASES	Attempt to deploy an unsigned image to staging; introduce a known-vulnerable dependency.
EXIT / DEFINITION OF DONE	Unsigned image rejected; SBOM attached to each release; HIGH/CRITICAL findings fail the build.
DEPENDENCIES	Task 16 (CI pipeline v1 with hard isolation gate), Task 17 (Infrastructure as code baseline)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.10 has a multi-stage Dockerfile (non-root uid 10001). Signing, SBOM and admission policy are described, not built.
PRIORITY / EFFORT	P1, effort S (1 day or less)

Workstream: Tenancy & Data

Task 22: Core schema and expand/contract migrations

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort L

OBJECTIVE	Define every table the product needs with tenancy built in.
WHY / ARCHITECTURE INTENT	The v2 model describes a generic fact store but not the coverage tracker, people list or job-posting records that the proven process produces; they cannot be bolted on later without churn.
IMPLEMENTATION	Alembic with expand then contract; tenants, memberships, accounts, account_runs, tool_calls, facts (append-only with superseded_at, visibility), fact_evidence, sources (tier, lineage group, content hash), wallets, wallet_ledger, alert_rules, alerts, audit_log, plus coverage_rows, people, job_postings, tech_detections, deliverables, tenant_profile; tenant_id NOT NULL with foreign key everywhere; insert-only grants on ledger and audit.
TESTING / FAILURE CASES	Migrate up and down on a scratch database; run the previous app release against the new schema; constraint tests (balance never negative, unique idempotency per tenant).
EXIT / DEFINITION OF DONE	Schema covers every requirement in the baseline; clean rebuild from migrations; backward-compatibility test passes; CODEOWNERS review recorded.
DEPENDENCIES	Task 18 (Managed Postgres, roles and extensions), Task 15 (Monorepo, tooling and local environment)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec Figure 7 gives the logical model and 0002_rls.sql creates account_runs, chunks and policies. Missing tables the PRD needs: lanes, coverage rows with proof, people, job postings, tech detections, deliverables, tenant profile.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 23: Forced RLS policies, fail-closed accessor and migration lint

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort M

OBJECTIVE	Make the database refuse cross-tenant access by construction.
WHY / ARCHITECTURE INTENT	Layers L3 to L6 must hold even if application code has a bug.
IMPLEMENTATION	Policy on every tenant table using (SELECT app_tenant()), FORCE ROW LEVEL SECURITY, WITH CHECK, NULLIF accessor; security_invoker views; reviewed list of SECURITY DEFINER functions; migration lint that fails the build when a tenant table lacks FORCE RLS or WITH CHECK or a role has BYPASSRLS.
TESTING / FAILURE CASES	Unset-context test, query-plan regression test, cross-tenant write test, definer-function inventory test.
EXIT / DEFINITION OF DONE	Lint passes on all tables; unset context returns zero rows; no unlisted definer function exists.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations)
STATUS	**PARTIAL**
STATUS EVIDENCE	migrations/0002_rls.sql (Spec 18.3) drafts the accessor, policies for account_runs, public_only for facts, checkpoint prefix policy and chunks. Not executed; migration lint not written.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 24: Tenant-scoped database access layer

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort S

OBJECTIVE	Give application code exactly one way to touch tenant data.
WHY / ARCHITECTURE INTENT	A bare SET LOCAL outside a transaction does nothing, which is how v1 would have leaked context through a pool.
IMPLEMENTATION	tenant_txn() as the only entry point; import lint forbidding direct session creation in src/; dedicated per-run connection variant for workers; pool sizing from the connection budget.
TESTING / FAILURE CASES	Pooled-leak test; 100 concurrent mixed-tenant requests over a small pool (reuse the spike harness).
EXIT / DEFINITION OF DONE	Lint enforced in CI; pooled-leak test and concurrency test pass.
DEPENDENCIES	Task 23 (Forced RLS policies, fail-closed accessor and migration lint)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.2 contains tenant_txn() using set_config(..., true) inside an explicit transaction. Compile-checked, not run.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 25: Isolation test suite as a hard CI gate

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort M

OBJECTIVE	Prove isolation on every build.
WHY / ARCHITECTURE INTENT	Zero tolerance: a single cross-tenant row is a release blocker.
IMPLEMENTATION	Extend the five drafted tests with planted canary vectors across partitions, role inventory (no BYPASSRLS, no owner), a tool-registry contract test proving no SQL, shell or arbitrary URL tool exists, and a query-plan regression test; run against the freshly migrated schema.
TESTING / FAILURE CASES	Mutation test: remove each policy in turn and confirm at least one test fails.
EXIT / DEFINITION OF DONE	Suite runs first in CI; every policy removal is detected; zero known gaps listed in the attack-scenario table.
DEPENDENCIES	Task 23 (Forced RLS policies, fail-closed accessor and migration lint), Task 24 (Tenant-scoped database access layer), Task 16 (CI pipeline v1 with hard isolation gate)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.9 drafts five tests (cross-tenant read, unset context, cross-tenant write, pool leak, checkpoint prefix). Missing: partition canaries, role inventory, tool-registry contract test, query-plan regression.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 26: Checkpoint isolation and LangGraph setup

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort S

OBJECTIVE	Isolate library-owned checkpoint tables that have no foreign key to runs.
WHY / ARCHITECTURE INTENT	v1 assumed a join that does not exist, leaving checkpoints unprotected.
IMPLEMENTATION	Prefix policy with WITH CHECK on checkpoints, checkpoint_blobs, checkpoint_writes; setup() run only by the migrator job; garbage collection of threads 7 days after a run becomes terminal; documented library pin.
TESTING / FAILURE CASES	Tenant A inserting a tenant B thread id gets InsufficientPrivilege; setup() is idempotent; GC removes only terminal threads.
EXIT / DEFINITION OF DONE	Prefix tests pass; GC job proven on fixtures; V11 closed against the pinned library version.
DEPENDENCIES	Task 23 (Forced RLS policies, fail-closed accessor and migration lint), Task 8 (Lock ADRs and the minimum viable infrastructure)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 4.3 and 18.3 set thread_id = tenant:run with a prefix policy on the three checkpoint tables; library behaviour (setup(), adelete_thread, autocommit, dict rows) is V11, unverified.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 27: Append-only audit log with archive

Phase 1: Concierge Service | Tenancy & Data | Priority P1 | Effort M

OBJECTIVE	Record who did what, tamper-evidently, per tenant.
WHY / ARCHITECTURE INTENT	Repudiation and support-access abuse are named STRIDE threats; clients also get an exportable audit trail.
IMPLEMENTATION	Insert-only table (actor, action, resource, time, hash of before and after state); 90 days queryable then 7 years in S3 with Object Lock; support access only through a logged, time-boxed grant from the tenant admin with reason code and a customer-visible entry.
TESTING / FAILURE CASES	UPDATE and DELETE attempts fail; hash-chain verification job; expired support grant stops working.
EXIT / DEFINITION OF DONE	Every state-changing endpoint writes an entry; archive proven to be immutable; tenant can export its own log.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 19 (Secrets, workload identity and per-tenant key management)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The audit_log table appears in the data model; the Object Lock archive and just-in-time support access are described in Spec 4.5 and the service design (section 3.1).
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 28: Authentication, RBAC and membership re-check

Phase 1: Concierge Service | Tenancy & Data | Priority P0 | Effort M

OBJECTIVE	Identify the caller and tenant from a verified token only.
WHY / ARCHITECTURE INTENT	Tenant must never come from the request body; membership must be re-checked so removed users lose access fast.
IMPLEMENTATION	Integrate the chosen IdP (SSO/SAML, MFA for admins); verify signature, issuer, audience, expiry; tenant and role from claims; role matrix from Spec 9.2 (admin, seller, viewer, later internal role); membership re-check with a 60 second cache; hashed, scoped API keys.
TESTING / FAILURE CASES	Forged tenant claim, expired and replayed tokens, removed user, role matrix test for every endpoint.
EXIT / DEFINITION OF DONE	All matrix cells tested; a removed user loses access within about a minute; no endpoint reads tenant id from the payload.
DEPENDENCIES	Task 24 (Tenant-scoped database access layer), Task 5 (Product-owner decisions on pricing, scope and policy)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.7 drafts JWT verification (RS256, issuer, audience, expiry) and role dependency. Provider is open (WorkOS or Clerk in the service design, generic IdP in the Spec).
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 29: Tenant profile and account configuration model

Phase 1: Concierge Service | Tenancy & Data | Priority P1 | Effort M

OBJECTIVE	Make lanes, rules and scope data, so each tenant gets the process it needs.
WHY / ARCHITECTURE INTENT	Today these settings live in the operator's head and the checklist; in a multi-client service they must be per-tenant configuration.
IMPLEMENTATION	Versioned JSON-schema-validated config: default lane set (Salesforce including field service, ServiceNow, SAP, Tableau and analytics, Veeva primary for life sciences and None elsewhere, Data and AI), proof library by industry and platform, writing rules (for example the Salesforce partner rule), people scope defaults (Director and above, Head, Senior Manager, global, security and IT compliance and IT audit excluded), cadence, alert rules, budgets, account owner. Changes audited.
TESTING / FAILURE CASES	Schema validation; an alert rule referencing another tenant's account is rejected; config change shows in the next run only.
EXIT / DEFINITION OF DONE	Profile can be created and edited by an admin; every run snapshots the profile it used; cross-tenant references impossible.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 28 (Authentication, RBAC and membership re-check)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The service design (2.1) describes the profile: who the client is, platform lanes, proof library, writing rules, account owner, people scope, cadence, alert rules, budgets. No schema or API exists.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Workstream: Deterministic Core

Task 30: Run lifecycle state machine

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort M

OBJECTIVE	Make run status changes safe under duplicate and concurrent events.
WHY / ARCHITECTURE INTENT	Every wallet settlement, cancellation and alert depends on a single, correct run status.
IMPLEMENTATION	Table-driven transition map for QUEUED, RUNNING, DISTILLING, VERIFYING, AWAITING_APPROVAL, COMPLETED, PARTIAL, FAILED, CANCELED, BUDGET_EXCEEDED; each transition is one UPDATE ... WHERE status = expected; terminal states trigger wallet settlement and checkpoint release; stuck-QUEUED detection beyond 60 s.
TESTING / FAILURE CASES	Property tests over random event orders; duplicate delivery; two workers racing the same transition; illegal transition attempts.
EXIT / DEFINITION OF DONE	No illegal transition reachable; duplicate events are no-ops; every terminal state has exactly one settlement trigger.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 24 (Tenant-scoped database access layer)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec Figure 4 defines the states and the optimistic-update rule; no code or table exists.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 31: Credit wallet ledger: reserve, commit, refund

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort L

OBJECTIVE	Guarantee no overspend, no double charge and a complete audit trail.
WHY / ARCHITECTURE INTENT	Ten parallel requests can each see a sufficient balance if debits happen later; reservation closes that hole.
IMPLEMENTATION	Entries reserve, commit, refund, topup (idempotent on provider event id), adjustment (reason code plus audit); BUDGET_EXCEEDED commits up to the cap and refunds the rest; full refund when the platform is at fault; nightly job checks balance equals ledger sum, no negative balance, exactly one settlement per terminal run.
TESTING / FAILURE CASES	10 parallel requests with funds for 3 yields 3 accepted and 7 returning 402; crash between steps; replayed webhooks; property tests. 100% coverage required on billing code.
EXIT / DEFINITION OF DONE	Invariant job green on a 10,000-run simulation; concurrency test passes; no ledger update or delete grant exists for the app role.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 24 (Tenant-scoped database access layer), Task 30 (Run lifecycle state machine)
STATUS	**PARTIAL**
STATUS EVIDENCE	Run creation code (Spec 18.8) already reserves credits atomically with a conditional UPDATE and writes a reserve ledger row. Missing: commit, refund, top-up, adjustment, settlement at terminal states and the nightly invariant job.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 32: Budget guard and circuit breakers

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort M

OBJECTIVE	Stop spend before the call is made, not after.
WHY / ARCHITECTURE INTENT	v1 observed spend after the fact. A guard inside the graph is the only control that prevents overspend mid-fan-out.
IMPLEMENTATION	Budget guard function called before every fan-out and paid call; per-run credits, broker spend, token cap per node, tool-call count, planner-loop count, wall clock; per-tenant daily cap at reservation time; per-broker Redis token bucket shared across workers; mapping from PRD credits to platform credits and categories.
TESTING / FAILURE CASES	Fan-out stops at the cap with status BUDGET_EXCEEDED and partial results; shared bucket enforced across two workers; cap change mid-run.
EXIT / DEFINITION OF DONE	No test run exceeds its cap by more than one in-flight call; breakdown by category reported at run end.
DEPENDENCIES	Task 31 (Credit wallet ledger: reserve, commit, refund), Task 30 (Run lifecycle state machine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 5.6 gives the control table (e.g. $5.00 broker cap per run, 60 tool calls, 2 planner loops, 4 h wall clock). PRD rule: one credit cap per run across all tools with a breakdown; no stopping to ask within the cap; ask before exceeding it.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 33: Tool-call ledger (exactly-once billing)

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort M

OBJECTIVE	Never pay twice for the same call within a run.
WHY / ARCHITECTURE INTENT	Crashes and slice resumes replay nodes; the ledger turns replays into free lookups and powers cost attribution.
IMPLEMENTATION	Unique (run_id, args_hash) with canonical JSON; responses over a size limit stored in S3 and referenced; cost_usd per row; views by tenant, run, account and broker; vendor idempotency keys where offered. Guarantee stated honestly: exactly-once billing, not exactly-once side effects, so one repeat per crash is the residual risk.
TESTING / FAILURE CASES	Kill between vendor call and insert; identical bytes on replay; argument ordering differences hash identically.
EXIT / DEFINITION OF DONE	Replay returns identical output; at most one repeated paid call per crash proven in chaos test.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 24 (Tenant-scoped database access layer)
STATUS	**PARTIAL**
STATUS EVIDENCE	ledgered_call() in Spec 18.4 hashes tool plus canonical arguments and replays stored responses. Missing: S3 offload of large responses, cost attribution views, vendor idempotency keys.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 34: Fact normalisation and fingerprint library

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort M

OBJECTIVE	Make equal facts collide so deltas and dedupe are exact.
WHY / ARCHITECTURE INTENT	If normalisation depends on a model, the same fact produces different fingerprints and every refresh looks like a change.
IMPLEMENTATION	Pure-function library with an attribute-key registry (for example exec.cto.name); title normaliser, currency and date parsers, domain canonicaliser, fingerprint function, helpers that always state a count together with its base.
TESTING / FAILURE CASES	Property tests with hypothesis for idempotence and collision behaviour; golden pairs from the nine accounts.
EXIT / DEFINITION OF DONE	Normalising twice equals normalising once; fingerprints stable across 1,000 random permutations; no model call anywhere in the package.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 6.3 specifies normalisation in code (case, titles, currency, dates, domain forms) and fingerprint = hash(entity, attribute, normalised value).
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 35: Evidence and grading engine

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort L

OBJECTIVE	Compute reproducible, calibratable grades from evidence.
WHY / ARCHITECTURE INTENT	A grade that merely sounds confident is a liability; the formula must be code that can be tested and tuned against the golden set.
IMPLEMENTATION	Pure functions: recency decay r = 0.5^(age/half-life), lineage-group maximum, confidence c = 1 - product(1 - w x r), grades A (c >= 0.85, two independent lineage groups, one T1 or T2), B (>= 0.70), C (>= 0.50), D, plus Conflicting and Blocked; weights in configuration; calibration report (predicted vs observed, expected calibration error).
TESTING / FAILURE CASES	Worked examples from the spec as unit tests (CTO name c = 0.975 grade A; two aggregators on one upstream c = 0.40 grade D); property tests on monotonicity and lineage handling.
EXIT / DEFINITION OF DONE	100% line coverage; worked examples pass; calibration error computed on golden v0 (target 0.05 or lower once the set is large enough).
DEPENDENCIES	Task 34 (Fact normalisation and fingerprint library)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started in code. Spec 6.2 fully specifies source tiers (T1 0.90, T2 0.75, T3 0.70, T4 0.40), half-lives (hiring 21 d, technology 90 d, leadership 180 d, funding 365 d), lineage groups, noisy-OR and the A-D rules with worked examples.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 36: Grade taxonomy reconciliation and prospect-safe rule

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort S

OBJECTIVE	Give clients one grading language backed by the numeric engine.
WHY / ARCHITECTURE INTENT	Two vocabularies produce contradictory files and break the rule that protects the first call from guesses.
IMPLEMENTATION	Define a ratified mapping, for example Confirmed = read in a primary (T1) source in this run; Probable = grade B or A without a primary read, reasoning stated; Hypothesis = system synthesis with a confirming question; Lead = secondary source only, aggregator labelled. Enforce as a computed field, a renderer rule and a lint: prospect-safe lists contain only Confirmed; absence of evidence is always Hypothesis.
TESTING / FAILURE CASES	Golden cases for each label; adversarial case where a Lead is copied into a prospect-safe section.
EXIT / DEFINITION OF DONE	Mapping ratified by the product owner; zero non-Confirmed items in prospect-safe lists across the golden set.
DEPENDENCIES	Task 35 (Evidence and grading engine), Task 4 (Source-document conflict register and resolution)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The PRD uses Confirmed, Probable, Hypothesis, Lead with the rule 'only Confirmed may be stated to a prospect'; Spec v2.0 uses A-D, Conflicting, Blocked and Unverified. The documents never map one to the other.
PRIORITY / EFFORT	P0, effort S (1 day or less)

Task 37: Temporal intelligence: shelf-life, debounce and freshness state machines

Phase 1: Concierge Service | Deterministic Core | Priority P1 | Effort M

OBJECTIVE	Convert noisy observations into stable, time-based facts.
WHY / ARCHITECTURE INTENT	Without hysteresis, one missed probe looks like a removed job posting and floods alerts; without event-time rules, late broker data rewrites history.
IMPLEMENTATION	Define event time (posting first and last seen, article date, detection dates, earnings date) vs processing time (fetch time). FSMs stored in database rows so they are checkpointable: signal shelf life FRESH -> AGING -> EXPIRED -> HISTORY; posting OPEN -> ABSENT_1 -> REMOVED (two probes) -> REOPENED; person ACTIVE -> QUIET -> RESUMED; leadership change confirmed only with two sources or one T1; refresh-due calculator per source cadence. Late or out-of-order data merges by event time with idempotent upserts and never moves first-seen later; a date range in a detection feed is a lead.
TESTING / FAILURE CASES	Jitter (one missed probe), short outage, late timestamps, restart mid-cycle, 18-month histories, clock skew, duplicate feed entries.
EXIT / DEFINITION OF DONE	Temporal facts match at least 20 scripted golden scenario timelines with 100% agreement; state survives process restart.
DEPENDENCIES	Task 35 (Evidence and grading engine), Task 30 (Run lifecycle state machine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started in code. The signals reference already assigns shelf lives (PRD and service design 5.1); Spec 6.4 gives debounce rules (a posting must be absent on two probes before REMOVED). No machine-readable state exists.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 38: Coverage tracker engine with proof

Phase 1: Concierge Service | Deterministic Core | Priority P0 | Effort M

OBJECTIVE	Make 'no silent omission' a database invariant.
WHY / ARCHITECTURE INTENT	The gate scripts catch structural defects but not a skipped source; the tracker with proof is the control for that.
IMPLEMENTATION	coverage_rows per account and run with status Done, Partial, Not done, Blocked, None; proof record (queries run, tool, result count); row definitions as tenant-editable data; tally computed in code; ordering by most valuable open work; rule that a run cannot reach COMPLETED while a High row lacks status and proof; 'lead' rule storing three differently worded queries across two tools.
TESTING / FAILURE CASES	Row without proof fails the gate; Blocked without reason rejected; tally always equals row count.
EXIT / DEFINITION OF DONE	All 33 rows representable; invariant enforced; tally recomputed from rows in 100% of test runs.
DEPENDENCIES	Task 22 (Core schema and expand/contract migrations), Task 30 (Run lifecycle state machine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: the 33-row tracker (PRD Appendix A: 5 Pass 1, 28 Pass 2; priorities High, Medium, Optional) is produced by hand in every Summary.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 39: Delta and change-detection engine

Phase 1: Concierge Service | Deterministic Core | Priority P1 | Effort M

OBJECTIVE	Produce an exact, traceable change record for each refresh.
WHY / ARCHITECTURE INTENT	Deltas feed both the Summary and the alert engine; every alert must trace to a graded, sourced row.
IMPLEMENTATION	Per-account delta record: new items, expired items moved to history, regraded items, count changes with their base; computed by fingerprint comparison between runs; stored with links to fact and evidence ids.
TESTING / FAILURE CASES	Replay two consecutive runs on fixtures; counts reconcile; identical inputs produce an empty delta; regrade and expiry cases.
EXIT / DEFINITION OF DONE	Delta output matches hand-computed expected deltas on 10 fixture pairs; idempotent.
DEPENDENCIES	Task 35 (Evidence and grading engine), Task 37 (Temporal intelligence: shelf-life, debounce and freshness state machines), Task 38 (Coverage tracker engine with proof)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: refresh was 'exercised once' manually (PRD 4.5); the new Summary opens with what changed.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 40: Job-posting classification and platform-sentence extraction

Phase 1: Concierge Service | Deterministic Core | Priority P1 | Effort M

OBJECTIVE	Classify postings deterministically before any model is involved.
WHY / ARCHITECTURE INTENT	A model classifying every posting is slow, costly and inconsistent; most cases can be decided by rules.
IMPLEMENTATION	Rule engine: extract the sentence containing a platform name; classify by title and responsibility verbs; send only ambiguous residue to the extractor tier; compute read-versus-matched counts; never classify as Build role from the title alone.
TESTING / FAILURE CASES	Labelled sample from the delivered accounts; precision and recall by class; adversarial postings (platform mentioned only as a user skill).
EXIT / DEFINITION OF DONE	Build-role precision at least 0.90 and recall at least 0.85 on the labelled sample; residue sent to the model under 25%.
DEPENDENCIES	Task 34 (Fact normalisation and fingerprint library)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started in code. Reference: rules proven in nine runs (keyword match is a lead; read the posting; classes Build role, Admin and owner role, Working tool, User role; counts read vs matched, per platform, open vs closed).
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 41: People list engine	
Phase 1: Concierge Service | Deterministic Core | Priority P1 | Effort M	
OBJECTIVE	Produce one deduplicated, scoped people list per account.
WHY / ARCHITECTURE INTENT	Overlapping people tables and re-pulled people waste credits and confuse sellers.
IMPLEMENTATION	Merge by profile URL first and name second; exclusions (security, IT compliance, IT audit); level, lane and priority assignment; never re-pull people already in a grid; counts of alumni, tenure and certifications computed in code and stated with their base.
TESTING / FAILURE CASES	Duplicate and renamed-person cases; exclusion list; counts match rows.
EXIT / DEFINITION OF DONE	Zero duplicates on fixtures; counts equal row totals; exclusions applied in 100% of cases.
DEPENDENCIES	Task 34 (Fact normalisation and fingerprint library), Task 22 (Core schema and expand/contract migrations)
STATUS	**PARTIAL**
STATUS EVIDENCE	People workbook logic exists as a reusable asset in the skill (PRD 4.1). Not integrated, not tested against the database.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Workstream: Broker Layer

Task 42: Broker adapter interface and credit metering	
Phase 1: Concierge Service | Broker Layer | Priority P0 | Effort M	
OBJECTIVE	Isolate brokers behind one contract with metering.
WHY / ARCHITECTURE INTENT	A broker can then be swapped or fail without touching the process, and every call is attributed to a tenant and run.
IMPLEMENTATION	Abstract adapter, common error taxonomy (rate-limited, out of credits, outage, bad input), metering hook writing to the tool ledger, normalised result shape, internal-only adapter names (never shown to clients).
TESTING / FAILURE CASES	Fake adapter test base; sum of metered cost equals ledger cost; error taxonomy mapped correctly.
EXIT / DEFINITION OF DONE	A new adapter passes the shared contract test suite without custom glue.
DEPENDENCIES	Task 33 (Tool-call ledger (exactly-once billing)), Task 32 (Budget guard and circuit breakers), Task 31 (Credit wallet ledger: reserve, commit, refund)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 4.4 defines one interface (search, fetch, jobs, news, detections, people by title, people by technology, profiles, posts, organizations, tech stack and teams).
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 43: Egress proxy, allow-lists and SSRF protection

Phase 1: Concierge Service | Broker Layer | Priority P0 | Effort M

OBJECTIVE	Control and log every outbound request.
WHY / ARCHITECTURE INTENT	Agents read untrusted pages; an unrestricted fetch is an SSRF and exfiltration path.
IMPLEMENTATION	Egress proxy with domain allow-list per adapter, blocking of link-local, private ranges and the instance metadata address, redirect cap, DNS rebinding defence, per-call logging with tenant and run; default-deny NetworkPolicies.
TESTING / FAILURE CASES	Attempts to reach 169.254.169.254, private ranges, redirect chains and rebinding hosts all fail and are logged.
EXIT / DEFINITION OF DONE	All SSRF test vectors blocked; every outbound call appears in logs with tenant and run.
DEPENDENCIES	Task 17 (Infrastructure as code baseline), Task 15 (Monorepo, tooling and local environment)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 11.2 and 14 require an allow-listed egress proxy, SSRF protection and default-deny network policy.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 44: Adapters: Parallel (search, fetch) and Firecrawl (scrape, monitor)

Phase 1: Concierge Service | Broker Layer | Priority P0 | Effort M

OBJECTIVE	Wrap the open-web and job-board brokers with the rules already learned.
WHY / ARCHITECTURE INTENT	Expensive modes used by mistake drain the cap; free readers return partial excerpts on job pages.
IMPLEMENTATION	Parallel search and fetch with fallback to built-in search; Firecrawl plain scrape only on job pages, browser pages, YouTube transcripts and change monitors; cost-guard that refuses expensive modes unless explicitly allowed; careers URL resolved by visiting, never constructed.
TESTING / FAILURE CASES	Cassette tests; cost-guard test for forbidden modes; rate-limit handling.
EXIT / DEFINITION OF DONE	Contract tests pass; no expensive mode reachable without explicit override.
DEPENDENCIES	Task 42 (Broker adapter interface and credit metering), Task 43 (Egress proxy, allow-lists and SSRF protection)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Routing and quirks are documented in the tools playbook (PRD 4.3): Parallel is free with one session per run and rate limits after a few fetches; Firecrawl costs one credit per plain scrape, five for JSON extraction, twenty or more for search with scrape.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 45: Adapter: PredictLeads (news, detections, job openings)

Phase 1: Concierge Service | Broker Layer | Priority P0 | Effort M

OBJECTIVE	Provide structured news, technology and posting data through the contract.
WHY / ARCHITECTURE INTENT	These are paid credits per call; pulling twice wastes the cap.
IMPLEMENTATION	News events with found-at from the last pull; technology detections with dates; open and closed job openings for 12 months with descriptions; responses persisted in the raw store; unopened feed items flagged 'Feed only'.
TESTING / FAILURE CASES	Cassette tests; second call in a run returns from the ledger; date-range detection recorded as lead.
EXIT / DEFINITION OF DONE	Second pull in the same run costs zero credits; fixtures replay to the same facts.
DEPENDENCIES	Task 42 (Broker adapter interface and credit metering), Task 43 (Egress proxy, allow-lists and SSRF protection)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference rules: pull once per run and save; structured news events; technology detections with first and last seen; 12 months of postings with descriptions.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 46: Adapters: BitScale and Sumble (people, organisations, tech stack)

Phase 1: Concierge Service | Broker Layer | Priority P1 | Effort L

OBJECTIVE	Cover people search and enrichment behind the contract.
WHY / ARCHITECTURE INTENT	People data is the costliest and most privacy-sensitive source group.
IMPLEMENTATION	Title search with lane and leadership variants and two seniority passes; association search only for gaps; enrichment with sample estimation and confirmation step; Sumble for organisations, tech stack and teams regardless; fill-rate and cost-per-usable-person metrics so the BitScale vs Sumble decision is data-based.
TESTING / FAILURE CASES	Paging on the same source; no re-adding of sources; hosted-file routing blocked; sample estimate before bulk enrichment.
EXIT / DEFINITION OF DONE	Fill rate and cost per usable senior person measured over 10 accounts; decision recorded.
DEPENDENCIES	Task 42 (Broker adapter interface and credit metering), Task 43 (Egress proxy, allow-lists and SSRF protection)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference rules: BitScale Find People by title (page on the same source), enrichment columns for profiles and posts (estimate on a sample first); Sumble people by technology, organisations, tech stack and teams (one credit per person plus per attribute; never route output to a hosted file). Whether BitScale alone covers people is an open decision.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 47: Broker resilience: retries, Blocked rows and pre-pulled files

Phase 1: Concierge Service | Broker Layer | Priority P0 | Effort M

OBJECTIVE	Turn broker failure into controlled partial results.
WHY / ARCHITECTURE INTENT	Today an empty balance halts the run; the platform must degrade rather than fail.
IMPLEMENTATION	Timeouts; 3 retries with exponential backoff and jitter; per-broker token buckets; wallet and broker-balance check before start; out-of-credit or outage marks affected coverage rows Blocked with reason and continues (run may finish PARTIAL); pre-pulled file ingestion path consumed instead of calling the broker.
TESTING / FAILURE CASES	Inject 429, 5xx, empty balance and timeouts; pre-pulled file yields the same downstream facts as a live pull.
EXIT / DEFINITION OF DONE	Run finishes PARTIAL with correct Blocked rows in every injected failure; no retry storms (one retry owner per failure class).
DEPENDENCIES	Task 44 (Adapters: Parallel (search, fetch) and Firecrawl (scrape, monitor)), Task 45 (Adapter: PredictLeads (news, detections, job openings)), Task 46 (Adapters: BitScale and Sumble (people, organisations, tech stack)), Task 38 (Coverage tracker engine with proof)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. PRD limitation: broker credit exhaustion stops a run and the workaround (pre-pulled files supplied by hand) is manual. Service design: outage or empty balance marks rows Blocked with a reason.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 48: Raw payload store and source-ID registry	
Phase 1: Concierge Service | Broker Layer | Priority P1 | Effort M	
OBJECTIVE	Persist raw responses once and keep source IDs stable.
WHY / ARCHITECTURE INTENT	Stable IDs make refreshes diff cleanly and keep the Evidence file consistent.
IMPLEMENTATION	S3 under tenant prefix with SSE-KMS; compact JSON with neutral file names; source table with tier, lineage group, content hash and permanent source ID; lifecycle rules to archive tier; presigned URLs short-lived and prefix-scoped.
TESTING / FAILURE CASES	Source IDs identical across refresh; presigned URL scope tests; lifecycle rule dry run.
EXIT / DEFINITION OF DONE	ID stability proven across two consecutive runs; no object readable outside its tenant prefix.
DEPENDENCIES	Task 42 (Broker adapter interface and credit metering), Task 19 (Secrets, workload identity and per-tenant key management)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: PRD 4.4 saves broker responses locally as compact JSON with stable source IDs reused across Summary, Evidence and refreshes; Spec 7.3 sets retention (30 days hot, 12 months archive).
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 49: Broker contract tests with cassettes and live smoke	
Phase 1: Concierge Service | Broker Layer | Priority P1 | Effort M	
OBJECTIVE	Detect broker schema drift before a customer does.
WHY / ARCHITECTURE INTENT	Brokers change response shapes; undetected drift silently breaks facts.
IMPLEMENTATION	Recorded cassettes per adapter; nightly live smoke with a spend cap per broker; schema comparison with alerts; fixtures reused by the eval harness.
TESTING / FAILURE CASES	Mutate a cassette field and confirm the drift check fails.
EXIT / DEFINITION OF DONE	Nightly job green for 7 days; drift alert proven.
DEPENDENCIES	Task 44 (Adapters: Parallel (search, fetch) and Firecrawl (scrape, monitor)), Task 45 (Adapter: PredictLeads (news, detections, job openings)), Task 46 (Adapters: BitScale and Sumble (people, organisations, tech stack)), Task 47 (Broker resilience: retries, Blocked rows and pre-pulled files), Task 16 (CI pipeline v1 with hard isolation gate)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 15 requires recorded cassettes plus a nightly live smoke with spend caps.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Workstream: Evaluation

Task 50: Evaluation harness and golden-set runner

Phase 1: Concierge Service | Evaluation | Priority P0 | Effort L

OBJECTIVE	Give every change a measurable baseline and a release gate.
WHY / ARCHITECTURE INTENT	Prompt, model and graph changes cannot be judged by eye; paired comparison needs a harness.
IMPLEMENTATION	Runner over golden fixtures (nine now, 50 at MVP, 200 later); judge model different from the generator, calibrated against human labels; rubric scoring with cited evidence; paired comparison with confidence intervals; CI eval-gate on main blocking promotion. Gates: precision of grade A at least 0.95 and B at least 0.85, coverage at least 0.80, citation validity at least 0.97, calibration error at most 0.05, cost within 10%, p95 duration within 15%.
TESTING / FAILURE CASES	Run the harness against a deliberately degraded prompt and confirm the gate blocks it.
EXIT / DEFINITION OF DONE	Harness runs end to end on golden v0 from replay fixtures; baseline metrics recorded; the gate blocks a seeded regression.
DEPENDENCIES	Task 9 (Golden set v0 from the nine delivered accounts), Task 35 (Evidence and grading engine), Task 16 (CI pipeline v1 with hard isolation gate)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 10.4 defines the golden set, judge hygiene, paired comparison and gates; Spec 17 says build it alongside the first graph.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 51: Adversarial and leakage corpus

Phase 1: Concierge Service | Evaluation | Priority P0 | Effort M

OBJECTIVE	Make prompt injection and leakage regressions impossible to ship.
WHY / ARCHITECTURE INTENT	Indirect injection from scraped pages is rated High likelihood and High impact in the risk register.
IMPLEMENTATION	Corpus of hostile pages (hidden instructions, tool-coaxing, requests to fetch other tenants' data), poisoned documents and planted canary strings; runner integrated in the eval gate; every real incident adds a case.
TESTING / FAILURE CASES	Each case must be dropped or sanitised with no tool misuse and an alert raised.
EXIT / DEFINITION OF DONE	100% pass for tool misuse and leakage canaries; the corpus has at least 50 cases at Phase 1 exit.
DEPENDENCIES	Task 50 (Evaluation harness and golden-set runner)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 10.4 requires a maintained corpus of injection pages and poisoned documents with a 100% pass rate.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Workstream: Agent Layer

Task 52: Model router, tier configuration and fallback

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort M

OBJECTIVE	Choose models by configuration, not code.
WHY / ARCHITECTURE INTENT	Models get deprecated and repriced; routing by tier makes upgrades a gated config change.
IMPLEMENTATION	Tier config file with dated IDs; per-step model names; fallback within tier before degrading; shared token bucket per model and provider account; jittered scheduling; degraded runs marked so graders and customers see it; change allowed only through the eval gate.
TESTING / FAILURE CASES	Simulated provider outage falls back; a config change that fails the eval gate is blocked.
EXIT / DEFINITION OF DONE	Swapping a tier model is a config change proven by an eval run; degraded flag visible in output.
DEPENDENCIES	Task 50 (Evaluation harness and golden-set runner), Task 32 (Budget guard and circuit breakers)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 10.1 defines tiers (planner, extractor, judge, embedding, rerank) with illustrative defaults claude-sonnet-5-5, claude-haiku-4-5-20251001 and claude-opus-5-5 marked [VERIFY]; service design 4.3 wants the model named per step.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 53: Prompt packaging, versioning and cache-aware layout

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort M

OBJECTIVE	Ship prompts and graph as one versioned artefact whose prefix caches reliably.
WHY / ARCHITECTURE INTENT	One changed byte in the prefix silently removes the discount; unversioned prompts corrupt in-flight runs.
IMPLEMENTATION	Prompts packaged with graph code and pinned in account_runs.graph_version; state schema_version with an upgrade function; prefix order tools, system and SOP, stable reference text, run-specific data, turns; tenant-free prefix; sorted tools and canonical JSON; explicit breakpoints; longer-lifetime option for static prefix with a warm-up call at slice start; telemetry for cache-read, cache-write and uncached tokens.
TESTING / FAILURE CASES	Byte-identical prefix across tenants; canary build injecting a timestamp must trip the cache-ratio alert within one hour.
EXIT / DEFINITION OF DONE	Cache-read ratio at least 70% on cacheable calls in staging; old graph versions resume on their own prompts.
DEPENDENCIES	Task 52 (Model router, tier configuration and fallback), Task 12 (Spike 3: prompt-cache ratio on real prompts)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Design only: Spec 10.2 gives the caching playbook; code snippets define GRAPH_VERSION and schema_version constants.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 54: Quarantine extractor for untrusted text

Phase 1: Concierge Service | Agent Layer | Priority P0 | Effort L

OBJECTIVE	Make indirect prompt injection structurally harmless.
WHY / ARCHITECTURE INTENT	Scraped pages, job posts and (later) private documents can contain instructions; nothing that reads them may hold a dangerous capability.
IMPLEMENTATION	Extractor tier with no tools; JSON-schema constrained output; quote spans for evidence; length caps; control-character stripping; cited URLs must resolve to the claimed domain; canary instruction detection (echo flags the source hostile and drops it); typed tool arguments with no tenant parameter anywhere.
TESTING / FAILURE CASES	Run the full adversarial corpus; malformed JSON; oversized output; homoglyph domains.
EXIT / DEFINITION OF DONE	100% pass on the adversarial corpus; no node that reads untrusted text can send, write or fetch.
DEPENDENCIES	Task 52 (Model router, tier configuration and fallback), Task 53 (Prompt packaging, versioning and cache-aware layout), Task 51 (Adversarial and leakage corpus)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 6.5 and ADR-005 define a tool-less extractor that emits schema-validated JSON; planner nodes never read raw text.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 55: Research graph skeleton

Phase 1: Concierge Service | Agent Layer | Priority P0 | Effort L

OBJECTIVE	Provide the orchestration skeleton where deterministic code decides and models fill slots.
WHY / ARCHITECTURE INTENT	The sample graph covers news, jobs, people and tech; the proven process has about 30 source types in seven groups.
IMPLEMENTATION	Nodes per Spec 6.1: baseline, plan, budget_guard, fetch_branch x N, quarantine_extract, merge, distill_grade, verify, render; branches extended to baseline and executive voice, news, jobs, tech stack, people, person-name searches, review sites and communities; node RetryPolicy of 3 attempts; BudgetExceeded mapped to status; maximum 2 gap-filling loops; checkpoint every superstep; facts reducer.
TESTING / FAILURE CASES	Fake adapters and fake models; resume after kill; branch failure marks facts Blocked and the run continues.
EXIT / DEFINITION OF DONE	Graph runs end to end on fixtures with deterministic fake models; every node idempotent through the ledger.
DEPENDENCIES	Task 26 (Checkpoint isolation and LangGraph setup), Task 33 (Tool-call ledger (exactly-once billing)), Task 32 (Budget guard and circuit breakers), Task 54 (Quarantine extractor for untrusted text), Task 35 (Evidence and grading engine), Task 38 (Coverage tracker engine with proof), Task 42 (Broker adapter interface and credit metering), Task 30 (Run lifecycle state machine), Task 34 (Fact normalisation and fingerprint library)
STATUS	**PARTIAL**
STATUS EVIDENCE	A reduced graph exists in Spec 18.4 (baseline, fan-out via Send, fetch_branch, distill) with four sample branches. Missing: plan, merge, verify, render nodes and the PRD's source groups.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 56: Time-sliced runner, Inngest slice loop and reconciler

Phase 1: Concierge Service | Agent Layer | Priority P0 | Effort L

OBJECTIVE	Run hours-long work as resumable short slices.
WHY / ARCHITECTURE INTENT	A multi-hour step inside one HTTP-triggered function is fragile; slicing plus the checkpoint makes any interruption safe.
IMPLEMENTATION	execute_slice with 240 s budget and a 60-slice (4 h) ceiling; dedicated per-run connection (autocommit, dict rows, prepare_threshold 0); Inngest function with per-tenant concurrency key, infrastructure retries only, 24 h approval wait, cancel check between supersteps; reconciler singleton with lease re-sending events for runs QUEUED beyond 60 s through a narrow definer function; finalizer settling the wallet.
TESTING / FAILURE CASES	Spike 2 scenarios in CI; Inngest outage for 10 minutes; duplicate event ids; cancel mid-slice.
EXIT / DEFINITION OF DONE	No run stuck beyond 5 minutes after recovery; every terminal run settled exactly once; V4 closed.
DEPENDENCIES	Task 55 (Research graph skeleton), Task 30 (Run lifecycle state machine)
STATUS	**PARTIAL**
STATUS EVIDENCE	Spec 18.5 (runner) and 18.6 (Inngest loop) are compile-checked listings, not executed; SDK call shapes are [VERIFY] (V4). Reconciler and finalizer are described but not coded.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 57: Human approval gates and credit-cap decision flow

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort M

OBJECTIVE	Put the operator's decisions into the workflow.
WHY / ARCHITECTURE INTENT	The cap proposal, people scope and approval to exceed a cap are the few moments a human matters; they must be explicit, audited and time-limited.
IMPLEMENTATION	Run proposal object (cap with breakdown by category, people scope, pursuing client); approve or reject events; Phase 1 operator approval before start; 24 h timeout cancels and refunds; request for spend beyond the cap pauses the run; all decisions audited.
TESTING / FAILURE CASES	Duplicate approvals idempotent; timeout settles correctly; rejected run refunds in full.
EXIT / DEFINITION OF DONE	Operator can approve or reject from the console; every decision has an audit entry.
DEPENDENCIES	Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 31 (Credit wallet ledger: reserve, commit, refund)
STATUS	**PARTIAL**
STATUS EVIDENCE	Interrupt and wait-for-event mechanics exist in the runner and orchestration listings. PRD decision points (credit cap with breakdown, people scope, pursuing client asked once) are not modelled.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 58: Agents: set-up, relationship history and company baseline

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort L

OBJECTIVE	Port the first passes of the proven process.
WHY / ARCHITECTURE INTENT	Pass 1 relationship rows depend on client systems that the platform will not hold until the vault exists.
IMPLEMENTATION	Domain resolution and careers-site verification by visiting; pursuing-client question asked once; relationship rows (prior customer status, alumni overlaps, vendor account team, key-contact profiles, basic fit) answered from operator-supplied inputs in Phase 1, otherwise marked 'waiting on the account owner' (vault connectors arrive in Phase 3); baseline from last three earnings releases, newsroom and leadership page; executive voice from last two earnings-call transcripts as focus, opportunity and challenge with the speaker named; every figure re-read in the release.
TESTING / FAILURE CASES	Fixture accounts; careers URL never constructed; figure-recheck test.
EXIT / DEFINITION OF DONE	Pass 1 and baseline rows reach Done, Partial or Blocked with proof on 5 fixture accounts.
DEPENDENCIES	Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 29 (Tenant profile and account configuration model), Task 44 (Adapters: Parallel (search, fetch) and Firecrawl (scrape, monitor)), Task 48 (Raw payload store and source-ID registry)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: PRD 4.2 passes 'Set up', 'Pass 1' and 'Company baseline' performed by hand in claude.ai.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 59: Agents: news, jobs and technology stack

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort L

OBJECTIVE	Port the news, hiring and technology evidence passes.
WHY / ARCHITECTURE INTENT	These are the highest-signal sources and the basis for refresh probes.
IMPLEMENTATION	News: company releases, structured feed, targeted journalist-phrased queries, tier-one items verified at source, unopened feed items marked 'Feed only'. Jobs: live board search once per lane keyword, then 12 months open and closed with the platform sentence read in every posting naming a platform and full text for every build role, classified by the deterministic engine. Tech: detections pulled once, lane-relevant rows kept with first and last seen, cross-checked against postings.
TESTING / FAILURE CASES	Fixture replay; counts of read vs matched; lead handling for date ranges.
EXIT / DEFINITION OF DONE	Rows reach status with proof on fixtures; classification counts match the labelled sample.
DEPENDENCIES	Task 58 (Agents: set-up, relationship history and company baseline), Task 45 (Adapter: PredictLeads (news, detections, job openings)), Task 40 (Job-posting classification and platform-sentence extraction), Task 37 (Temporal intelligence: shelf-life, debounce and freshness state machines), Task 48 (Raw payload store and source-ID registry)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: PRD 4.2 (News, Jobs, Tech stack) run sequentially by hand; the platform runs them in parallel.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 60: Agents: people, person-name searches, review sites and communities

Phase 1: Concierge Service | Agent Layer | Priority P1 | Effort L

OBJECTIVE	Port the people evidence passes with their ordering rule.
WHY / ARCHITECTURE INTENT	Person-name searches depend on the people list existing; running them first wastes credits.
IMPLEMENTATION	Title search on the domain with two seniority passes; association search only for gaps (IT, engineering, product); people named in research with profile URLs; full profiles for every senior person; posts for priority A only; person-name searches (speaking past and upcoming, podcasts, YouTube, Medium, Substack, vendor customer stories) for the top ten; review sites and vendor communities checked and recorded as None when empty.
TESTING / FAILURE CASES	Ordering test (person searches wait for the people step); duplicate-pull test; credit-cap test.
EXIT / DEFINITION OF DONE	People list and person-name rows complete with proof on fixtures; zero re-pulled people.
DEPENDENCIES	Task 58 (Agents: set-up, relationship history and company baseline), Task 46 (Adapters: BitScale and Sumble (people, organisations, tech stack)), Task 41 (People list engine), Task 48 (Raw payload store and source-ID registry)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: PRD 4.2 people and person-name passes.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 61: Verification node: Leads, citations and contradictions

Phase 1: Concierge Service | Agent Layer | Priority P0 | Effort M

OBJECTIVE	Stop weak claims from reaching the brief.
WHY / ARCHITECTURE INTENT	A confident guess presented as fact is the most common cause of a bad first call.
IMPLEMENTATION	Re-query loop for each Lead (three phrasings, two tools); parked leads re-searched when later evidence appears; judge tier different from the generator checks that the cited span supports each claim and flags contradictions; unsupported claims dropped.
TESTING / FAILURE CASES	Seeded false claims and contradictory sources; citation precision measured.
EXIT / DEFINITION OF DONE	Citation validity at least 0.97 on golden v0; every Lead either upgraded with proof or labelled as Lead.
DEPENDENCIES	Task 59 (Agents: news, jobs and technology stack), Task 60 (Agents: people, person-name searches, review sites and communities), Task 35 (Evidence and grading engine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference rules: a Lead needs three differently worded queries across two tools; absence of evidence is a Hypothesis; Spec 6.1 adds a judge-tier citation and contradiction check.
PRIORITY / EFFORT	P0, effort M (2-3 days)

Task 62: Distillation and writing-rule engine

Phase 1: Concierge Service | Agent Layer | Priority P0 | Effort L

OBJECTIVE	Turn graded evidence into readable, safe output.
WHY / ARCHITECTURE INTENT	Writing rules are the second source of client-visible defects; they must be enforced by code, not remembered by a model.
IMPLEMENTATION	Model drafts narrative, hypotheses each with a confirming question, first-call questions and what-changes-a-decision; deterministic lints apply tenant writing rules (plain English headings, abbreviations written out on first use, internal program names explained, sensitive findings (breach, layoff, lawsuit) as preparation never openers, Salesforce-partner framing rule, banned dash styles); every statement carries a source ID and a matching graded row.
TESTING / FAILURE CASES	Lint corpus of known-bad sentences; statement-to-row matching test; sensitive-finding placement test.
EXIT / DEFINITION OF DONE	Zero lint violations and zero unmatched statements on golden v0 outputs.
DEPENDENCIES	Task 61 (Verification node: Leads, citations and contradictions), Task 36 (Grade taxonomy reconciliation and prospect-safe rule), Task 38 (Coverage tracker engine with proof)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: PRD distill step and writing rules (plain English, abbreviations spelled out, sensitive findings as preparation, Salesforce-partner rule).
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 63: Model-split experiment per workflow step

Phase 1: Concierge Service | Agent Layer | Priority P2 | Effort M

OBJECTIVE	Decide model use per step from measured gate-failure rates.
WHY / ARCHITECTURE INTENT	Over-paying on pull steps or under-powering distill both hurt margin or quality.
IMPLEMENTATION	A/B runs per step with the router config; measure gate failure rate, citation validity, cost and duration per step and model on at least 20 runs.
TESTING / FAILURE CASES	Paired comparison with confidence intervals through the eval harness.
EXIT / DEFINITION OF DONE	Decision record with numbers; router defaults updated through the eval gate.
DEPENDENCIES	Task 62 (Distillation and writing-rule engine), Task 50 (Evaluation harness and golden-set runner)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. PRD lists the split as undecided; service design 4.3 proposes the stronger model on distill and build and testing a faster model on pull steps.
PRIORITY / EFFORT	P2, effort M (2-3 days)

Workstream: Deliverables & Gate

Task 64: Document builder: Account Summary and Sources and Evidence

Phase 1: Concierge Service | Deliverables & Gate | Priority P1 | Effort L

OBJECTIVE	Generate the Word deliverables from stored account state.
WHY / ARCHITECTURE INTENT	Output consistency is a product feature; builds must come from data, not from chat.
IMPLEMENTATION	Container wrapping the existing builder; Account Summary in the PRD order (plain narrative first, caveats, graded tables by lane, key people, Findings from LinkedIn, executive voice, hypotheses with confirming questions, first-call questions, coverage tracker sorted by value); Sources and Evidence with stable source IDs, link, date, grade, exact sentence relied on; page size and margins per spec; Word to PDF conversion for checking.
TESTING / FAILURE CASES	Golden render comparison; no table row lost at a page break; long-table stress test.
EXIT / DEFINITION OF DONE	Builds identical output from identical state; PDF check shows zero lost rows on 5 fixtures.
DEPENDENCIES	Task 62 (Distillation and writing-rule engine), Task 29 (Tenant profile and account configuration model)
STATUS	**PARTIAL**
STATUS EVIDENCE	The Node docx builder library exists in the skill assets and produces the same look every time. Not containerised and not fed from database state.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 65: Workbook builders: People List, Job Postings, Tech Stack and Teams

Phase 1: Concierge Service | Deliverables & Gate | Priority P1 | Effort M

OBJECTIVE	Generate the three standing workbooks from stored state.
WHY / ARCHITECTURE INTENT	Sellers work in these files; formula errors or mismatched totals erode trust.
IMPLEMENTATION	People_List with Summary sheet (exclusions and counts); Job_Postings (12 months open and closed, class, platform sentence, full description, link); Tech_Stack_and_Teams (detections and team structure cross-checked with dates); recalculation with zero formula errors; totals equal row counts.
TESTING / FAILURE CASES	Recalculate and scan every cell; compare totals with database counts.
EXIT / DEFINITION OF DONE	Zero formula errors; totals equal row counts on fixtures.
DEPENDENCIES	Task 41 (People list engine), Task 40 (Job-posting classification and platform-sentence extraction), Task 64 (Document builder: Account Summary and Sources and Evidence)
STATUS	**PARTIAL**
STATUS EVIDENCE	Python openpyxl logic for the people and jobs workbooks exists as a reusable asset. Not integrated; Tech Stack and Teams builder status not evidenced.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 66: Workspace package and Internal-Only file split

Phase 1: Concierge Service | Deliverables & Gate | Priority P1 | Effort M

OBJECTIVE	Package the client-shareable set and keep internal material out of it.
WHY / ARCHITECTURE INTENT	The two-output rule starts here: anything that names client staff or comes from calls must never enter the share or the zip.
IMPLEMENTATION	Workspace zip builder; separate Internal-Only Word file generated only when needed and stored outside the share path; redelivery ships the complete set with one line on what changed; neutral raw-data names.
TESTING / FAILURE CASES	Zip contents scan; Internal-Only file never present in share or zip across 20 builds.
EXIT / DEFINITION OF DONE	Zip passes the same scans as the client files; Internal-Only file isolated by path and by role.
DEPENDENCIES	Task 64 (Document builder: Account Summary and Sources and Evidence), Task 65 (Workbook builders: People List, Job Postings, Tech Stack and Teams)
STATUS	**PARTIAL**
STATUS EVIDENCE	Packaging logic exists in the skill (RESTORE.md, deliverables, scripts, raw data with neutral names). The internal-only document is called 'Rajesh Only' in the PRD and is produced by hand.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 67: Gate service: pre-delivery checks and loop-back

Phase 1: Concierge Service | Deliverables & Gate | Priority P0 | Effort L

OBJECTIVE	Block delivery of any defective file set.
WHY / ARCHITECTURE INTENT	The gate is what maintains quality when nobody is watching; it catches structural defects (dropped rows, orphan citations, banned dashes, broker names).
IMPLEMENTATION	Containerise the four scripts; run on the rendered PDF and on the unzipped package XML, relationship files, workbook cells and properties; checks: zero orphan citations, banned dashes, stale phrases, excluded names, broker names or credit counts; tracker tally recomputed from rows; narrative reread against tables; new tenant-leakage scan; on failure loop back to the failing step with the defect attached up to a limit, then route to an operator; per-step gate-failure metric.
TESTING / FAILURE CASES	Seed each defect type and confirm it is caught; confirm the gate does not catch a skipped source (that is the tracker's job).
EXIT / DEFINITION OF DONE	Every seeded defect blocked; loop-back limit proven; gate failure rate reported per step.
DEPENDENCIES	Task 64 (Document builder: Account Summary and Sources and Evidence), Task 65 (Workbook builders: People List, Job Postings, Tech Stack and Teams), Task 66 (Workspace package and Internal-Only file split), Task 51 (Adversarial and leakage corpus)
STATUS	**PARTIAL**
STATUS EVIDENCE	rowcheck.py, doc_checks.py, third_party_scan.py and strip_tool_names.py exist and run today. Not containerised; no loop-back policy; no tenant leakage scan.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 68: Public renderer role and leak detector (Phase A)	
Phase 1: Concierge Service | Deliverables & Gate | Priority P1 | Effort M	
OBJECTIVE	Guarantee the client-facing deliverable uses public data only.
WHY / ARCHITECTURE INTENT	Private data must not reach a public file even if a prompt fails.
IMPLEMENTATION	render_public switches to the public_reader role in its transaction; leak detector checks outputs for any private-visibility identifiers and canary strings; Phase 3 extends it with n-gram and embedding similarity.
TESTING / FAILURE CASES	Attempt to read private rows as public_reader; plant a private canary and confirm it never appears.
EXIT / DEFINITION OF DONE	Zero private rows readable by the public role; canary never leaks in 100 builds.
DEPENDENCIES	Task 23 (Forced RLS policies, fail-closed accessor and migration lint), Task 64 (Document builder: Account Summary and Sources and Evidence), Task 67 (Gate service: pre-delivery checks and loop-back)
STATUS	**PARTIAL**
STATUS EVIDENCE	public_only policy and the public_reader role are drafted in 0002_rls.sql; chunks have a CHECK fixing visibility to private. Renderer and detector not built.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Workstream: API & Apps

Task 69: Core API v1	
Phase 1: Concierge Service | API & Apps | Priority P0 | Effort L	
OBJECTIVE	Expose runs, accounts, briefs, facts, changes and wallet over a stable API.
WHY / ARCHITECTURE INTENT	Run creation returns 202; research is asynchronous; p99 must stay under 500 ms.
IMPLEMENTATION	FastAPI with path versioning; Idempotency-Key enforced by unique constraint; ETag on run status; cursor pagination; problem-details errors; token bucket per tenant and per user with endpoint cost weights; OpenAPI published; reconciler path for failed event send.
TESTING / FAILURE CASES	OpenAPI contract tests; role matrix; load test at 3x forecast; replayed idempotency keys.
EXIT / DEFINITION OF DONE	All endpoints in the catalogue implemented for Phase 1 scope; p99 under 500 ms at 3x forecast; no tenant id in any payload.
DEPENDENCIES	Task 28 (Authentication, RBAC and membership re-check), Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 31 (Credit wallet ledger: reserve, commit, refund), Task 29 (Tenant profile and account configuration model), Task 67 (Gate service: pre-delivery checks and loop-back), Task 57 (Human approval gates and credit-cap decision flow)
STATUS	**PARTIAL**
STATUS EVIDENCE	POST /v1/runs is drafted in Spec 18.8 (idempotent, reserved, outbox-safe) and the endpoint catalogue is in Spec 9.1. The rest is not built.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 70: Operator console

Phase 1: Concierge Service | API & Apps | Priority P1 | Effort L

OBJECTIVE	Let 6th operate concierge runs across tenants.
WHY / ARCHITECTURE INTENT	Phase 1 is concierge: operators approve runs, handle gate failures and maintain the checklist.
IMPLEMENTATION	Queue of runs with approve or reject; spend by tenant, account and broker (internal view); versioned editor for checklist, lessons and signals reference applying to every tenant's next run; gate-failure triage; just-in-time support access with reason code and audit.
TESTING / FAILURE CASES	Role test (operator only); edit-then-run test proving the new rule applies; audit entries present.
EXIT / DEFINITION OF DONE	Operator can run a full concierge cycle without database access.
DEPENDENCIES	Task 69 (Core API v1), Task 57 (Human approval gates and credit-cap decision flow), Task 32 (Budget guard and circuit breakers)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. FR-16 and service design 2.2 define the operator console: run approvals, spend per tenant and account, process-rule maintenance.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 71: Client portal v1

Phase 1: Concierge Service | API & Apps | Priority P1 | Effort L

OBJECTIVE	Give clients results, evidence, downloads and alert settings.
WHY / ARCHITECTURE INTENT	Phase 1 clients see results, alerts and schedules through a portal even though 6th triggers runs.
IMPLEMENTATION	Next.js portal: brief viewer with evidence drill-down (grade, confidence, citations), coverage tracker view, downloads of Word and Excel, alert rule settings, budget approval, cost per run in credits and categories (never broker names), optional delivery to a connected Drive or SharePoint folder.
TESTING / FAILURE CASES	Playwright flow sign-in to brief to download; HTML scan proves no broker or credit-vendor name.
EXIT / DEFINITION OF DONE	Critical paths green in staging; zero broker names in rendered output.
DEPENDENCIES	Task 69 (Core API v1), Task 67 (Gate service: pre-delivery checks and loop-back)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 4.1 proposes Next.js with server components for tenant scoping; Spec lists the portal in the container view.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Workstream: Scheduler & Alerts

Task 72: Scheduler: cadences, probes and escalation

Phase 1: Concierge Service | Scheduler & Alerts | Priority P1 | Effort L

OBJECTIVE	Keep accounts fresh at a fraction of full-run cost.
WHY / ARCHITECTURE INTENT	Refresh is the value of the service; Spec estimates about $1.14 expected refresh cost vs $5.60 for a full run at 15% escalation (illustrative).
IMPLEMENTATION	Cadence per source type: live job board weekly; job openings, technology and speaking monthly; news and leadership weekly; earnings quarterly on the reporting calendar; senior people, priority A profiles and review sites quarterly; posts monthly only for people with recent activity. Probe cheap high-signal sources; escalate on delta score or maximum age; jitter across the hour; per-tenant concurrency keys; refresh starts from the last tracker, reuses source IDs, re-runs High rows and expired signals, moves expired signals to history; wallet check before start.
TESTING / FAILURE CASES	Skip and alert when the standing budget is low; probe thresholds on fixtures; thundering-herd test.
EXIT / DEFINITION OF DONE	A month of scheduled refreshes on fixtures delivers on time; measured refresh cost recorded against the 80% saving assumption.
DEPENDENCIES	Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 37 (Temporal intelligence: shelf-life, debounce and freshness state machines), Task 39 (Delta and change-detection engine), Task 32 (Budget guard and circuit breakers), Task 29 (Tenant profile and account configuration model)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Reference: manual refresh exercised once (PRD 4.5); cadence table in service design 5.1; probe-then-escalate in Spec 6.4.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 73: Alert engine v1: rules, dedupe, digest and email

Phase 1: Concierge Service | Scheduler & Alerts | Priority P1 | Effort L

OBJECTIVE	Deliver changes that matter without alert fatigue.
WHY / ARCHITECTURE INTENT	Alerts are the visible value of refresh; noisy alerts are muted and the product with them.
IMPLEMENTATION	Rules: new build or admin role in a tenant lane, technology or commercial leadership change, new technology detection, tier-one news after source verification, speaker slots, priority A person posting after a quiet period, Lead regraded, budget alerts; dedupe key (rule, account, fingerprint); leadership needs two sources or one T1; postings absent on two probes before REMOVED; per-account daily cap; digest batching above a threshold; sensitive signals worded as preparation; email delivery and weekly digest.
TESTING / FAILURE CASES	Replay deltas; duplicate suppression; digest threshold; wording lint for sensitive signals.
EXIT / DEFINITION OF DONE	Alert latency within the 24 h target on fixtures; zero duplicate alerts; every alert traces to a graded row.
DEPENDENCIES	Task 39 (Delta and change-detection engine), Task 72 (Scheduler: cadences, probes and escalation), Task 69 (Core API v1)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 5.3 gives the rule table; Spec 6.4 adds noise controls. Scheduled monitoring and alerts are a stated non-goal of the current system.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Workstream: Release & Operations

Task 74: Observability baseline

Phase 1: Concierge Service | Release & Operations | Priority P1 | Effort L

OBJECTIVE	See every run, call and cost.
WHY / ARCHITECTURE INTENT	Cost per run per tenant is a product metric; isolation violations must be visible immediately.
IMPLEMENTATION	OpenTelemetry across API, workers and jobs; spans per node and per tool or model call (tier, model, tokens, cache tokens, cost, outcome); Langfuse self-hosted for prompts; metrics by plan tier, region and tool (tenant dimension only in traces and logs to avoid cardinality blow-up); canary strings to prove prompts and documents never reach logs.
TESTING / FAILURE CASES	Log scanner for planted canaries; trace continuity across slices; cardinality check.
EXIT / DEFINITION OF DONE	One trace per run across all slices; zero canary leaks in logs; dashboards for the full metrics catalogue.
DEPENDENCIES	Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 69 (Core API v1), Task 20 (Kubernetes platform and autoscaling)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 13 defines OpenTelemetry, run id as trace id, self-hosted Langfuse, structured logs with a field allow-list and a metrics catalogue.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 75: SLOs, alerting and runbooks

Phase 1: Concierge Service | Release & Operations | Priority P1 | Effort M

OBJECTIVE	Define what good looks like and who is paged.
WHY / ARCHITECTURE INTENT	Honest targets (99.9% availability, p99 under 500 ms, 95% run success, cache ratio at least 70%) need alerts and written responses.
IMPLEMENTATION	SLO definitions and burn-rate pages (14.4x over 1 h and 6x over 6 h); page for isolation violations, replication lag over 60 s, run-success burn and worker crash loops; ticket and digest tiers; a runbook for every page; blameless post-incident template.
TESTING / FAILURE CASES	Fire each alert in staging and walk the runbook.
EXIT / DEFINITION OF DONE	Every page has a tested runbook; on-call rota in place.
DEPENDENCIES	Task 74 (Observability baseline)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 13.1 and 13.4 define SLOs, burn-rate alerting and severity levels.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 76: Load and chaos test suite

Phase 1: Concierge Service | Release & Operations | Priority P1 | Effort L

OBJECTIVE	Prove durability and scaling claims with experiments.
WHY / ARCHITECTURE INTENT	'No lost run, no duplicate paid call' is a promise that must be demonstrated.
IMPLEMENTATION	Kill worker mid-slice; block the workflow engine for 10 minutes; force database failover; terminate most agent nodes; serve a poisoned page; inject a timestamp into a cache prefix; k6 load for the API and a synthetic run injector with stub brokers at 3x forecast.
TESTING / FAILURE CASES	Pass criteria as listed in the spec.
EXIT / DEFINITION OF DONE	All six experiments pass; SLOs met at 3x forecast.
DEPENDENCIES	Task 56 (Time-sliced runner, Inngest slice loop and reconciler), Task 74 (Observability baseline), Task 20 (Kubernetes platform and autoscaling), Task 49 (Broker contract tests with cassettes and live smoke)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 15.1 lists six experiments with pass criteria.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 77: Security review and hardening (Phase 1 scope)

Phase 1: Concierge Service | Release & Operations | Priority P1 | Effort M

OBJECTIVE	Find security defects before paying tenants do.
WHY / ARCHITECTURE INTENT	An external RLS and prompt-injection assessment is required before the first enterprise customer.
IMPLEMENTATION	STRIDE walkthrough; verify each OWASP LLM control; SAST, dependency audit, container scan, DAST; RLS bypass attempts; secrets-in-logs and secrets-in-prompts checks; injection red-team; commission a focused external assessment.
TESTING / FAILURE CASES	Findings triaged by severity with retest.
EXIT / DEFINITION OF DONE	No high or critical finding open; external assessment report on file.
DEPENDENCIES	Task 25 (Isolation test suite as a hard CI gate), Task 54 (Quarantine extractor for untrusted text), Task 67 (Gate service: pre-delivery checks and loop-back), Task 51 (Adversarial and leakage corpus), Task 69 (Core API v1), Task 68 (Public renderer role and leak detector (Phase A))
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 14 provides the STRIDE table, OWASP LLM Top 10 mapping (2025 numbering, V9 unverified) and compliance programme.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 78: Parity replay of the nine delivered accounts

Phase 1: Concierge Service | Release & Operations | Priority P0 | Effort L

OBJECTIVE	Prove the platform reproduces the proven manual output.
WHY / ARCHITECTURE INTENT	Moving from one operator and one model to parallel agents risks quality loss that structural checks do not see.
IMPLEMENTATION	Replay the nine accounts using recorded fixtures plus a live subset (three accounts within an agreed cap); compare coverage row statuses and proof, grade distribution, people counts, job counts read vs matched, gate result, credits vs cap and wall-clock time (manual: one working day; Spec target: p50 under 10 minutes).
TESTING / FAILURE CASES	Side-by-side diff of Summary tables; reviewer assessment of narrative quality.
EXIT / DEFINITION OF DONE	At least 95% of High rows match status; zero silent omissions; zero gate-escaped defects; cost within the cap.
DEPENDENCIES	Task 67 (Gate service: pre-delivery checks and loop-back), Task 50 (Evaluation harness and golden-set runner), Task 9 (Golden set v0 from the nine delivered accounts), Task 60 (Agents: people, person-name searches, review sites and communities), Task 62 (Distillation and writing-rule engine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The nine delivered file sets are the baseline; no replay exists.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 79: Production deployment with canary and rollback

Phase 1: Concierge Service | Release & Operations | Priority P1 | Effort L

OBJECTIVE	Release safely, including while multi-hour runs are in flight.
WHY / ARCHITECTURE INTENT	Blue-green swaps ignore long-running runs and their state format.
IMPLEMENTATION	Argo Rollouts canary with analysis on error rate, p99, run-failure ratio, cache-read ratio and cost per run; graph versions pinned per run (old and new workers coexist until old has no active runs); expand/contract migrations by the migrator job; same image digest from dev to staging to prod in separate accounts; smoke tests with recorded fixtures; rollback by routing change plus version flag.
TESTING / FAILURE CASES	Bad release triggers automatic rollback; in-flight run completes on its original graph version.
EXIT / DEFINITION OF DONE	Rollback under 1 hour proven; first production release completed through the canary path.
DEPENDENCIES	Task 20 (Kubernetes platform and autoscaling), Task 21 (Container build, signing and supply chain), Task 77 (Security review and hardening (Phase 1 scope)), Task 78 (Parity replay of the nine delivered accounts), Task 74 (Observability baseline), Task 76 (Load and chaos test suite)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 12 defines canary 5/25/100 with auto-rollback and graph version pinning.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Workstream: Docs & Handover

Task 80: Technical documentation and runbooks

Phase 1: Concierge Service | Docs & Handover | Priority P1 | Effort M

OBJECTIVE	Make the system operable by someone else.
WHY / ARCHITECTURE INTENT	The solo or small-team risk is single-person knowledge; a fresh reader must be able to run it.
IMPLEMENTATION	Architecture overview, ADR log, setup guide, deployment and rollback runbooks, failover steps, stuck-run recovery, wallet reconciliation, key rotation, API reference from OpenAPI, troubleshooting guide, procedure for changing process rules.
TESTING / FAILURE CASES	A developer not involved in the build clones, runs, deploys to staging and recovers a stuck run using only the docs.
EXIT / DEFINITION OF DONE	Cold-start test passed by an independent developer.
DEPENDENCIES	Task 79 (Production deployment with canary and rollback)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec chapters are design documents, not operating documentation.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 81: Client and operator documentation

Phase 1: Concierge Service | Docs & Handover | Priority P1 | Effort M

OBJECTIVE	Let clients and operators use the product without developer help.
WHY / ARCHITECTURE INTENT	Grading labels and the prospect-safe rule must be understood by sellers to be useful.
IMPLEMENTATION	Client quick start; portal guide; alert guide; grading explained (Confirmed, Probable, Hypothesis, Lead and what may be said to a prospect); limitations; support path; operator guide; sub-processor and data-handling summary.
TESTING / FAILURE CASES	Usability test with two non-technical users completing the primary workflow unaided.
EXIT / DEFINITION OF DONE	Both testers finish the primary workflow without assistance.
DEPENDENCIES	Task 71 (Client portal v1), Task 70 (Operator console)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Workstream: Release & Operations

Task 82: Concierge pilot launch with two paying tenants Phase 1: Concierge Service | Release & Operations | Priority P0 | Effort L	
OBJECTIVE	Prove the service with real clients before building self-serve.
WHY / ARCHITECTURE INTENT	The process is proven; tenant isolation and scheduling are what need proving next.
IMPLEMENTATION	Onboard two tenants; 6th reviews the first run per tenant; invoice per account plus credit pass-through tracked in the wallet; weekly digest; one month of scheduled refresh; written position on two tenants researching one account; instrument the seller-usefulness metric; track isolation defects, gate failure rate and cost per run per tenant.
TESTING / FAILURE CASES	Weekly pilot review against the metrics; incident log.
EXIT / DEFINITION OF DONE	Two paying tenants live; zero isolation defects; refresh delivered on schedule for one month; grade calibration measured on at least 100 accounts across runs.
DEPENDENCIES	Task 79 (Production deployment with canary and rollback), Task 80 (Technical documentation and runbooks), Task 81 (Client and operator documentation), Task 73 (Alert engine v1: rules, dedupe, digest and email), Task 72 (Scheduler: cadences, probes and escalation), Task 71 (Client portal v1), Task 70 (Operator console), Task 6 (Commercial and legal clearance (broker terms, privacy law, DPA)), Task 75 (SLOs, alerting and runbooks)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Phase 1 exit criteria differ: service design says two paying tenants, zero isolation defects and a month of on-time scheduled refresh; Spec says operators run 100 accounts end to end with grade calibration measured.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Phase 2: Self-Serve

Exit milestone: Runs start without an operator; ten paying tenants; SLOs met for 30 days; DR rehearsal passed.

Workstream: Self-Serve

Task 83: Self-serve onboarding and policy-based auto-start Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort L	
OBJECTIVE	Let clients sign up, add accounts and start runs without an operator.
WHY / ARCHITECTURE INTENT	Concierge does not scale; automation must stay inside budget and scope policy.
IMPLEMENTATION	Sign-up, organisation creation, SSO invite; tenant profile wizard; account list import; lane selection; policy engine for auto-start (budget, scope, daily cap); automatic retry then routing to an operator after repeated gate failures.
TESTING / FAILURE CASES	Policy edge cases (budget exhausted, scope outside policy); abuse tests for sign-up.
EXIT / DEFINITION OF DONE	A run starts without an operator; gate failure rate below the agreed threshold.
DEPENDENCIES	Task 82 (Concierge pilot launch with two paying tenants), Task 29 (Tenant profile and account configuration model), Task 31 (Credit wallet ledger: reserve, commit, refund), Task 32 (Budget guard and circuit breakers)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. FR-01 and service design phase two: client-triggered runs within policy, account onboarding, lane selection.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 84: Billing and metering with Stripe

Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort L

OBJECTIVE	Charge correctly and show usage clearly.
WHY / ARCHITECTURE INTENT	Free runs with real broker costs lose money fastest.
IMPLEMENTATION	Stripe usage-based billing; platform fee per active account per month plus metered credits; top-ups through checkout with a webhook idempotent on event id; invoices; usage reports by category (people, postings, news, pages read) without vendor names; reconciliation of wallet against provider invoices.
TESTING / FAILURE CASES	Duplicate and out-of-order webhooks; failed payment; nightly invariants.
EXIT / DEFINITION OF DONE	Invoices match ledger totals for 3 test tenants; zero double charges.
DEPENDENCIES	Task 31 (Credit wallet ledger: reserve, commit, refund), Task 82 (Concierge pilot launch with two paying tenants)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 9.4 and service design section 6 define the wallet, platform fee per active account and metered credits.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 85: Custom alert rules and Slack, Teams and webhook delivery

Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort L

OBJECTIVE	Let tenants shape alerts and receive them where they work.
WHY / ARCHITECTURE INTENT	Alert value depends on fit to each seller's workflow.
IMPLEMENTATION	Rules over any graded field, lane, grade or count threshold scoped to the tenant; Slack and Teams apps; signed webhooks (HMAC-SHA256 over timestamp and body, five-minute tolerance, retries for 24 h, event ids); digest batching; rules cannot reference another tenant's account.
TESTING / FAILURE CASES	Signature verification; replay attacks; cross-tenant reference attempts.
EXIT / DEFINITION OF DONE	All delivery channels proven; webhook signatures verified by a reference client.
DEPENDENCIES	Task 73 (Alert engine v1: rules, dedupe, digest and email), Task 82 (Concierge pilot launch with two paying tenants)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. FR-09 and service design 2.1.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 86: Plain-language question answering over tenant facts

Phase 2: Self-Serve | Self-Serve | Priority P2 | Effort L

OBJECTIVE	Answer questions across a tenant's accounts with cited evidence.
WHY / ARCHITECTURE INTENT	Sellers ask cross-account questions that no single brief answers.
IMPLEMENTATION	Retrieval scoped by RLS before retrieval; pgvector with tenant partitions; hybrid lexical plus vector retrieval; answers tagged Public (with link) or Internal (document and date), with internal passages gated by role (internal source class activated in Phase 3).
TESTING / FAILURE CASES	Cross-tenant canary queries; citation precision; refusal of out-of-scope queries.
EXIT / DEFINITION OF DONE	Citation precision at least 0.95 on a question set; zero cross-tenant results.
DEPENDENCIES	Task 82 (Concierge pilot launch with two paying tenants), Task 23 (Forced RLS policies, fail-closed accessor and migration lint), Task 50 (Evaluation harness and golden-set runner)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 2.1 and 4.1 describe questions scoped to the tenant's own accounts.
PRIORITY / EFFORT	P2, effort L (4-7 days)

Task 87: Tenant data governance: export, deletion, offboarding and audit viewer

Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort L

OBJECTIVE	Give tenants control of their data and a trail of who touched it.
WHY / ARCHITECTURE INTENT	Enterprise buyers ask before they upload anything; deletion must be provable.
IMPLEMENTATION	Data export; cascade deletion of derived data, raw broker results and embeddings with client confirmation (public cache entries are not tenant data); offboarding that schedules KMS key deletion after the retention window; audit viewer with filters by actor and resource and export; DSAR workflow.
TESTING / FAILURE CASES	Delete a test tenant and scan every table and bucket prefix; backup restore drill after key deletion.
EXIT / DEFINITION OF DONE	Zero rows or objects remain after deletion; key deletion renders backups unreadable in the drill.
DEPENDENCIES	Task 27 (Append-only audit log with archive), Task 28 (Authentication, RBAC and membership re-check), Task 19 (Secrets, workload identity and per-tenant key management), Task 82 (Concierge pilot launch with two paying tenants)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. FR-14 (export, deletion, crypto-shredding) and FR-15 (audit viewer).
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 88: Warm-standby region and disaster-recovery game-day

Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort XL

OBJECTIVE	Survive loss of the primary region within stated targets.
WHY / ARCHITECTURE INTENT	The spec corrects v1: replication is asynchronous, so the honest claim is a target proven by rehearsal.
IMPLEMENTATION	Aurora Global secondary in the standby region, S3 cross-region replication with replica KMS key, replicated secrets, pre-pulled images, Route 53 failover; scripted seven-step failover runbook; wallet reconciliation against provider invoices after failover; confirm workflow-engine failover and event-id dedupe across regions.
TESTING / FAILURE CASES	Quarterly game-day with a timer; verify the isolation suite against the new writer.
EXIT / DEFINITION OF DONE	RPO and RTO measured against targets in a rehearsal; runbook scripted; result recorded.
DEPENDENCIES	Task 82 (Concierge pilot launch with two paying tenants), Task 17 (Infrastructure as code baseline), Task 18 (Managed Postgres, roles and extensions)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 11.3 sets RPO under 60 s and RTO 30 min as targets with async Aurora Global Database; support in ap-south-2 and latency are unverified (V2).
PRIORITY / EFFORT	P1, effort XL (over 7 days)

Task 89: Unit-economics and cost optimisation pass

Phase 2: Self-Serve | Self-Serve | Priority P2 | Effort M

OBJECTIVE	Reach at least 70% gross margin at list price on measured numbers.
WHY / ARCHITECTURE INTENT	Brokers, not tokens, dominate; optimising the wrong lever wastes time.
IMPLEMENTATION	Replace assumptions with ledger and telemetry data; tune probe thresholds; skip attributes still inside their half-life; measure cache-read ratio and spot savings; negotiate volume pricing using ledger spend; review reserved database capacity.
TESTING / FAILURE CASES	Cost per run compared against baseline within 10%.
EXIT / DEFINITION OF DONE	Model recomputed from real data; margin at list price at or above 70% or a repricing proposal approved.
DEPENDENCIES	Task 82 (Concierge pilot launch with two paying tenants), Task 13 (Spike 4: broker economics on 20 real accounts), Task 74 (Observability baseline), Task 63 (Model-split experiment per workflow step)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 16 is an illustrative parametrised model: about $7.04 per run at 1,000 runs per month falling to $4.68 at 100,000, with brokers 64-70% of cost.
PRIORITY / EFFORT	P2, effort M (2-3 days)

Task 90: SOC 2 evidence collection

Phase 2: Self-Serve | Self-Serve | Priority P1 | Effort M

OBJECTIVE	Be ready when enterprise clients ask for a report.
WHY / ARCHITECTURE INTENT	Enterprise clients will ask before uploading call recordings.
IMPLEMENTATION	Choose tooling (Vanta, Drata or manual first); map controls to evidence from Terraform, CI, audit log and access reviews; policies; vendor register; incident process.
TESTING / FAILURE CASES	Internal readiness assessment before auditor engagement.
EXIT / DEFINITION OF DONE	Evidence collected continuously; Type I audit scheduled or issued.
DEPENDENCIES	Task 27 (Append-only audit log with archive), Task 77 (Security review and hardening (Phase 1 scope)), Task 17 (Infrastructure as code baseline), Task 82 (Concierge pilot launch with two paying tenants)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 14.4 requires evidence collection from MVP, Type I around month 6 and Type II after the observation window.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 91: Optional shared public cache	
Phase 2: Self-Serve | Self-Serve | Priority P3 | Effort M	
OBJECTIVE	Avoid paying twice for the same public data where allowed.
WHY / ARCHITECTURE INTENT	Savings exist only when accounts overlap, and some clients will want a contractual promise that nothing is shared.
IMPLEMENTATION	Cache keyed by account domain and source type with freshness window; stores only raw public broker output (never derived or internal data); tenant setting; tenants never see hit rates or anything revealing another tenant's interest; a copy is kept under the requesting tenant.
TESTING / FAILURE CASES	Leakage tests for derived data and for hit-rate exposure; licence check.
EXIT / DEFINITION OF DONE	Cache returns only raw public data; no tenant-visible cache metrics; legal cleared.
DEPENDENCIES	Task 48 (Raw payload store and source-ID registry), Task 82 (Concierge pilot launch with two paying tenants), Task 6 (Commercial and legal clearance (broker terms, privacy law, DPA))
STATUS	NOT STARTED
STATUS EVIDENCE	Not started. Service design 3.3 defines a per-tenant setting, default off for enterprise tenants; broker licences may limit sharing.
PRIORITY / EFFORT	P3, effort M (2-3 days)

Phase 3: Enhanced (Private Vault)

Exit milestone: Leakage canaries at zero; retrieval recall targets met; one tenant in production on internal material with a completed security review; SOC 2 Type I issued or scheduled.

Workstream: Private Vault

Task 92: Per-account vault, tenant keys and bridge and silo tiers	
Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort L	
OBJECTIVE	Create the secure home for private client material.
WHY / ARCHITECTURE INTENT	Private material is the highest-risk data class; it should land on a platform whose isolation has already run in production.
IMPLEMENTATION	Per-account vault inside the tenant with its own access rules; per-tenant customer-managed KMS key; Bridge tier (dedicated vector partitions and key) and Silo tier (dedicated Aurora cluster, optional node pool); retention policy per tenant; regional storage option in line with the residency decision.
TESTING / FAILURE CASES	Extend the isolation suite for vault tables; tier migration drill; user with access to one account cannot read another's vault.
EXIT / DEFINITION OF DONE	Isolation suite green with vault tables; tier upgrade performed on a test tenant without downtime.
DEPENDENCIES	Task 87 (Tenant data governance: export, deletion, offboarding and audit viewer), Task 19 (Secrets, workload identity and per-tenant key management), Task 18 (Managed Postgres, roles and extensions), Task 82 (Concierge pilot launch with two paying tenants)
STATUS	NOT STARTED
STATUS EVIDENCE	Not started. Spec 4.1 defines Pooled, Bridge and Silo tiers; service design 7.6 adds per-account vaults and regional options.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 93: Internal-source connectors

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort XL

OBJECTIVE	Bring a client's own material into its vault with authorisation.
WHY / ARCHITECTURE INTENT	Nothing may be ingested without a tenant user authorising the source.
IMPLEMENTATION	Build only connectors the first customer needs, starting with Grain (already in use) and a CRM export; OAuth tokens encrypted with the tenant key context and never logged; per-user email consent or forward-to-ingest address; account mapping by domain and client tag with client confirmation when ambiguous.
TESTING / FAILURE CASES	Token revocation; mapping ambiguity; rate limits; consent withdrawal.
EXIT / DEFINITION OF DONE	Chosen connectors ingest sample data end to end with logged authorisation.
DEPENDENCIES	Task 92 (Per-account vault, tenant keys and bridge and silo tiers), Task 5 (Product-owner decisions on pricing, scope and policy)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 7.2 lists call platforms (Grain, Gong, Zoom, Teams, Meet), email (Gmail, Microsoft 365), CRM (Salesforce, HubSpot, Dynamics), documents (Drive, SharePoint, OneDrive) and notes. Which ones the first Phase B customer needs is an open question.
PRIORITY / EFFORT	P1, effort XL (over 7 days)

Task 94: Ingestion pipeline: scan, parse and transcribe

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort L

OBJECTIVE	Turn files and recordings into clean text safely.
WHY / ARCHITECTURE INTENT	Uploads are untrusted; large files must not hurt API latency.
IMPLEMENTATION	Direct-to-S3 multipart upload; quarantine bucket, virus scan, type and size checks before promotion; parser (Unstructured or LlamaParse) in a sandboxed namespace with no network except the vendor; transcription with speaker labels (Deepgram or AssemblyAI) when only audio exists; classification (type, date, account mapping, sensitivity); audio kept only if the client asks.
TESTING / FAILURE CASES	Malware sample (EICAR), malformed PDFs, very long calls.
EXIT / DEFINITION OF DONE	Nothing parsed before the scan passes; transcripts carry speaker labels on a sample of real calls.
DEPENDENCIES	Task 92 (Per-account vault, tenant keys and bridge and silo tiers), Task 93 (Internal-source connectors), Task 43 (Egress proxy, allow-lists and SSRF protection)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 8.1 defines presigned multipart upload, quarantine bucket with virus scan, parsers in a sandboxed pod, speech recognition with diarization (V6 unverified).
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 95: Redaction policy tiers and redaction maps

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort M

OBJECTIVE	Protect sensitive data while keeping business contacts useful.
WHY / ARCHITECTURE INTENT	Redacting names from CRM notes destroys their value; blanket rules are wrong.
IMPLEMENTATION	Presidio plus custom recognisers; per-tenant tier; map of detected entities stored; originals kept in the vault; reversal only for the internal role.
TESTING / FAILURE CASES	Planted PII corpus; Standard tier keeps business contact names, titles and work emails but removes government IDs, cards, health data and credentials.
EXIT / DEFINITION OF DONE	Recall on planted PII at or above 0.95 for tier-removed classes.
DEPENDENCIES	Task 94 (Ingestion pipeline: scan, parse and transcribe)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 8.2 defines Strict, Standard (default) and Minimal tiers; service design 7.3 adds a stored redaction map with originals preserved.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 96: Chunking, embeddings and hybrid retrieval

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort L

OBJECTIVE	Retrieve the right private passages for a tenant and account, fast and safely.
WHY / ARCHITECTURE INTENT	Filtered ANN can under-return for small tenants; embeddings alone miss names and acronyms.
IMPLEMENTATION	Documents: 400-800 tokens with 10-15% overlap and heading path; transcripts: 60-90 s speaker-turn windows; CRM notes one per chunk; embeddings via the model's 1536-dimension option stored as halfvec; hybrid vector top-50 plus full-text top-50 with reciprocal rank fusion, rerank, top 8 with document id and span.
TESTING / FAILURE CASES	Per-tenant-size recall probes; canary vectors; deletion propagation within 24 h.
EXIT / DEFINITION OF DONE	Recall and nDCG targets met per size band; zero leakage canaries.
DEPENDENCIES	Task 94 (Ingestion pipeline: scan, parse and transcribe), Task 95 (Redaction policy tiers and redaction maps), Task 23 (Forced RLS policies, fail-closed accessor and migration lint), Task 86 (Plain-language question answering over tenant facts)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Spec 7.2 and 8.3 specify halfvec(1536), 16 hash partitions, HNSW with iterative scan, structure-aware chunking and hybrid fusion; V3 (pgvector features) unverified.
PRIORITY / EFFORT	P1, effort L (4-7 days)

Task 97: Internal fact extraction and relationship state

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P1 | Effort M

OBJECTIVE	Use the client's own knowledge to complete the research.
WHY / ARCHITECTURE INTENT	Pass 1 rows are often Blocked or Partial today; the vault turns them into Done with proof.
IMPLEMENTATION	Extraction inside the quarantine boundary (platforms, incumbent partners, programs, budget cycles, objections, decisions, next steps, people and roles); grade Internal linked to the passage; never Confirmed from internal material alone unless the client marks the source authoritative; relationship state per person (met, emailed, presented to, no contact, with dates).
TESTING / FAILURE CASES	Planted contradictory internal and public facts; grade-promotion rules.
EXIT / DEFINITION OF DONE	Pass 1 rows reach Done with proof on a pilot tenant; no internal-only fact graded Confirmed.
DEPENDENCIES	Task 96 (Chunking, embeddings and hybrid retrieval), Task 54 (Quarantine extractor for untrusted text), Task 58 (Agents: set-up, relationship history and company baseline), Task 41 (People list engine)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Service design 7.3 and 7.4 describe candidate facts graded Internal, Pass 1 answered from the vault, hypotheses tested against internal evidence and relationship state per person.
PRIORITY / EFFORT	P1, effort M (2-3 days)

Task 98: Two-output rule: Internal Only view and leakage scan

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P0 | Effort L

OBJECTIVE	Guarantee private material never appears in a client-facing file.
WHY / ARCHITECTURE INTENT	This is the product's trust promise; a single leak ends enterprise sales.
IMPLEMENTATION	Client-facing set uses public evidence only, with internal material only in neutral form (Client, call notes with a date, email by date and subject, no links); Internal Only view in the portal and as a separate Word file marked on cover and header; internal_reader role plus RBAC (seller, admin); leak detector with n-gram and embedding similarity against private chunks, redaction-map scan and internal source IDs, blocking delivery; /brief/internal never exported.
TESTING / FAILURE CASES	Planted private canaries and adversarial prompts; viewer role denied; zip and shared-folder scans.
EXIT / DEFINITION OF DONE	Leakage canaries at zero across the adversarial corpus and 100 builds.
DEPENDENCIES	Task 97 (Internal fact extraction and relationship state), Task 68 (Public renderer role and leak detector (Phase A)), Task 67 (Gate service: pre-delivery checks and loop-back)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started in full. Phase A pieces exist (public role policy, public renderer). Design: Spec chapter 8 Figure 8 and service design 7.5.
PRIORITY / EFFORT	P0, effort L (4-7 days)

Task 99: Enhanced-version security review and production pilot

Phase 3: Enhanced (Private Vault) | Private Vault | Priority P0 | Effort L

OBJECTIVE	Release the enhanced version to one real tenant safely.
WHY / ARCHITECTURE INTENT	Private recordings and contracts raise the stakes of every earlier control.
IMPLEMENTATION	External penetration test focused on RLS and prompt injection with private documents; signed DPA and sub-processor list; zero-retention and no-training verified contractually; deletion propagation within 24 h verified; one production tenant using internal material.
TESTING / FAILURE CASES	Retest of all findings; deletion drill on the pilot tenant.
EXIT / DEFINITION OF DONE	No open high or critical findings; both source documents' exit criteria met.
DEPENDENCIES	Task 98 (Two-output rule: Internal Only view and leakage scan), Task 90 (SOC 2 evidence collection), Task 88 (Warm-standby region and disaster-recovery game-day)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. Phase 3 exit criteria: leakage canaries at zero, retrieval recall targets met, SOC 2 Type I issued or scheduled (Spec) and one tenant using internal material in production with a completed security review (service design).
PRIORITY / EFFORT	P0, effort L (4-7 days)

Workstream: Docs & Handover

Task 100: Handover, maintenance plan and 30/60/90-day plan

Phase 3: Enhanced (Private Vault) | Docs & Handover | Priority P2 | Effort M

OBJECTIVE	Close the programme cleanly and keep the system maintainable.
WHY / ARCHITECTURE INTENT	Single-person knowledge and unmanaged debt are the long-term risks.
IMPLEMENTATION	Access and asset inventory (cloud, domains, repositories, secret locations without secrets, licences, broker accounts); known-issues backlog with severity and next action; maintenance cadence and backup/restore checks; 30/60/90-day plan (stabilise, improve, scale and iterate); lessons learned; release tag; confirm Open Door can read account state and alerts without any outreach capability built in.
TESTING / FAILURE CASES	Independent developer follows the inventory to regain access in a drill.
EXIT / DEFINITION OF DONE	Closure checklist complete; inventory verified in a recovery drill.
DEPENDENCIES	Task 99 (Enhanced-version security review and production pilot), Task 80 (Technical documentation and runbooks), Task 81 (Client and operator documentation)
STATUS	**NOT STARTED**
STATUS EVIDENCE	Not started. The generic roadmap supplies the closure pattern; the service design leaves Open Door as a separate module reading the same account state.
PRIORITY / EFFORT	P2, effort M (2-3 days)

6. Completion dashboard

Status counts

Status	Tasks	Share	
DONE	1	1%	
PARTIAL	24	24%	
NOT STARTED	75	75%	

Is any task finished? Only Task 1 (documentation of the current system). 24 tasks have usable drafts (mostly spec code that has not been run). The other 75 need to be built from scratch.

Top 10 next tasks to execute

Rank	Task	Title	Why now
1	T3	Unified requirements baseline with REQ-IDs	Unblocks every later task by giving each requirement an ID, an owner and a test; closes the gap between the PRD and the v2 functional list.
2	T4	Source-document conflict register and resolution	Nine contradictions across the documents must be settled before architecture is locked; cheap to do, expensive to defer.
3	T5	Product-owner decisions on pricing, scope and policy	Pricing, residency, people-data scope and the two-tenant policy change the data model and billing.
4	T6	Commercial and legal clearance (broker terms, privacy law, DPA)	Broker resale terms and privacy law can invalidate the cost model or the people-data scope; needs counsel lead time.
5	T7	Close the verification register (V1-V11)	Eleven unverified vendor facts sit under design decisions; closing them is mostly reading and small smoke tests.
6	T8	Lock ADRs and the minimum viable infrastructure	Fixes orchestrator, runtime, hosting and auth choices so the spikes test the real stack.
7	T9	Golden set v0 from the nine delivered accounts	Creates the baseline from the nine delivered accounts; the evaluation harness and parity replay both depend on it.
8	T10	Spike 1: tenant isolation with forced RLS and pooled connections	Isolation is the hardest property to retrofit and the product's trust promise.
9	T11	Spike 2: durable slices, resume and the tool ledger	Proves runs survive kills without double billing; decides whether Temporal is needed.
10	T13	Spike 4: broker economics on 20 real accounts	Brokers are 64-70% of cost; measure cost per run on 20 accounts before committing months of build.

Tasks 3 to 7 and 9 can run in parallel as soon as their prerequisites close; the three spikes need the ADR task first.

Top 5 project risks and mitigations

Risk	Likelihood / impact	Mitigation	Where it is handled
Cross-tenant data leakage	Low likelihood, severe impact	Forced RLS, fail-closed policies, isolation suite as a hard CI gate, canary vectors, external RLS and injection assessment before paying tenants.	RLS, tenant access layer, isolation suite, secrets and keys, Phase 1 security review
Broker cost or licence terms break the economics	High likelihood, high impact	Broker-economics spike on 20 accounts, per-run caps, probe-then-escalate refresh, legal review of resale terms, volume pricing from ledger data.	Legal clearance, broker economics spike, budget guard, scheduler, cost pass
Scope and team mismatch	High likelihood, high impact	Re-baseline after Phase 0; choose the minimum viable infrastructure; defer DR and Kubernetes complexity until the pilot; keep private vault last.	ADR task, Phase 0 gate, Kubernetes and DR tasks
Quality drops when parallel agents replace the operator	Medium likelihood, high impact	Deterministic core first, coverage tracker with proof, gate service, evaluation harness, parity replay of nine accounts.	Coverage tracker, gate, evaluation harness, parity replay
Prompt injection and privacy exposure	High likelihood, high impact	Quarantine extractor, typed tools, no SQL or shell tools, adversarial corpus at 100% pass, two-output rule with leak detector, counsel review of people data.	Quarantine extractor, adversarial corpus, public renderer, two-output rule

Full task table

Task	Title	Workstream	Phase	Status	Prio	Effort	Depends on
1	Business case, PRD and current-system documentation	Decisions & Spikes	0	DONE	P0	S	None
2	Service design and Architecture Specification v2.0 review	Decisions & Spikes	0	PARTIAL	P0	M	1
3	Unified requirements baseline with REQ-IDs	Decisions & Spikes	0	PARTIAL	P0	M	1, 2
4	Source-document conflict register and resolution	Decisions & Spikes	0	NOT STARTED	P0	S	3
5	Product-owner decisions on pricing, scope and policy	Decisions & Spikes	0	NOT STARTED	P0	S	4
6	Commercial and legal clearance (broker terms, privacy law, DPA)	Decisions & Spikes	0	NOT STARTED	P0	M	5
7	Close the verification register (V1-V11)	Decisions & Spikes	0	NOT STARTED	P0	M	2
8	Lock ADRs and the minimum viable infrastructure	Decisions & Spikes	0	PARTIAL	P0	M	4, 5, 7
9	Golden set v0 from the nine delivered accounts	Decisions & Spikes	0	NOT STARTED	P0	M	1
10	Spike 1: tenant isolation with forced RLS and pooled connections	Decisions & Spikes	0	NOT STARTED	P0	L	8
11	Spike 2: durable slices, resume and the tool ledger	Decisions & Spikes	0	NOT STARTED	P0	L	8
12	Spike 3: prompt-cache ratio on real prompts	Decisions & Spikes	0	NOT STARTED	P1	M	8, 7
13	Spike 4: broker economics on 20 real accounts	Decisions & Spikes	0	NOT STARTED	P0	L	8, 6, 9
14	Phase 0 gate review and design amendment	Decisions & Spikes	0	NOT STARTED	P0	S	10, 11, 12, 13
15	Monorepo, tooling and local environment	Foundations	1	PARTIAL	P0	S	14
16	CI pipeline v1 with hard isolation gate	Foundations	1	PARTIAL	P0	S	15
17	Infrastructure as code baseline	Foundations	1	NOT STARTED	P0	L	8, 15
18	Managed Postgres, roles and extensions	Foundations	1	NOT STARTED	P0	M	17
19	Secrets, workload identity and per-tenant key management	Foundations	1	NOT STARTED	P1	M	17
20	Kubernetes platform and autoscaling	Foundations	1	NOT STARTED	P1	L	17, 15
21	Container build, signing and supply chain	Foundations	1	PARTIAL	P1	S	16, 17
22	Core schema and expand/contract migrations	Tenancy & Data	1	PARTIAL	P0	L	18, 15
23	Forced RLS policies, fail-closed accessor and migration lint	Tenancy & Data	1	PARTIAL	P0	M	22
24	Tenant-scoped database access layer	Tenancy & Data	1	PARTIAL	P0	S	23
25	Isolation test suite as a hard CI gate	Tenancy & Data	1	PARTIAL	P0	M	23, 24, 16
26	Checkpoint isolation and LangGraph setup	Tenancy & Data	1	PARTIAL	P0	S	23, 8
27	Append-only audit log with archive	Tenancy & Data	1	NOT STARTED	P1	M	22, 19
28	Authentication, RBAC and membership re-check	Tenancy & Data	1	PARTIAL	P0	M	24, 5
29	Tenant profile and account configuration model	Tenancy & Data	1	NOT STARTED	P1	M	22, 28

Task	Title	Workstream	Phase	Status	Prio	Effort	Depends on
30	Run lifecycle state machine	Deterministic Core	1	NOT STARTED	P0	M	22, 24
31	Credit wallet ledger: reserve, commit, refund	Deterministic Core	1	PARTIAL	P0	L	22, 24, 30
32	Budget guard and circuit breakers	Deterministic Core	1	NOT STARTED	P0	M	31, 30
33	Tool-call ledger (exactly-once billing)	Deterministic Core	1	PARTIAL	P0	M	22, 24
34	Fact normalisation and fingerprint library	Deterministic Core	1	NOT STARTED	P0	M	22
35	Evidence and grading engine	Deterministic Core	1	NOT STARTED	P0	L	34
36	Grade taxonomy reconciliation and prospect-safe rule	Deterministic Core	1	NOT STARTED	P0	S	35, 4
37	Temporal intelligence: shelf-life, debounce and freshness state machines	Deterministic Core	1	NOT STARTED	P1	M	35, 30
38	Coverage tracker engine with proof	Deterministic Core	1	NOT STARTED	P0	M	22, 30
39	Delta and change-detection engine	Deterministic Core	1	NOT STARTED	P1	M	35, 37, 38
40	Job-posting classification and platform-sentence extraction	Deterministic Core	1	NOT STARTED	P1	M	34
41	People list engine	Deterministic Core	1	PARTIAL	P1	M	34, 22
42	Broker adapter interface and credit metering	Broker Layer	1	NOT STARTED	P0	M	33, 32, 31
43	Egress proxy, allow-lists and SSRF protection	Broker Layer	1	NOT STARTED	P0	M	17, 15
44	Adapters: Parallel (search, fetch) and Firecrawl (scrape, monitor)	Broker Layer	1	NOT STARTED	P0	M	42, 43
45	Adapter: PredictLeads (news, detections, job openings)	Broker Layer	1	NOT STARTED	P0	M	42, 43
46	Adapters: BitScale and Sumble (people, organisations, tech stack)	Broker Layer	1	NOT STARTED	P1	L	42, 43
47	Broker resilience: retries, Blocked rows and pre-pulled files	Broker Layer	1	NOT STARTED	P0	M	44, 45, 46, 38
48	Raw payload store and source-ID registry	Broker Layer	1	NOT STARTED	P1	M	42, 19
49	Broker contract tests with cassettes and live smoke	Broker Layer	1	NOT STARTED	P1	M	44, 45, 46, 47, 16
50	Evaluation harness and golden-set runner	Evaluation	1	NOT STARTED	P0	L	9, 35, 16
51	Adversarial and leakage corpus	Evaluation	1	NOT STARTED	P0	M	50
52	Model router, tier configuration and fallback	Agent Layer	1	NOT STARTED	P1	M	50, 32
53	Prompt packaging, versioning and cache-aware layout	Agent Layer	1	NOT STARTED	P1	M	52, 12
54	Quarantine extractor for untrusted text	Agent Layer	1	NOT STARTED	P0	L	52, 53, 51
55	Research graph skeleton	Agent Layer	1	PARTIAL	P0	L	26, 33, 32, 54, 35, 38, 42, 30, 34
56	Time-sliced runner, Inngest slice loop and reconciler	Agent Layer	1	PARTIAL	P0	L	55, 30
57	Human approval gates and credit-cap decision flow	Agent Layer	1	PARTIAL	P1	M	56, 31

Task	Title	Workstream	Phase	Status	Prio	Effort	Depends on
58	Agents: set-up, relationship history and company baseline	Agent Layer	1	NOT STARTED	P1	L	56, 29, 44, 48
59	Agents: news, jobs and technology stack	Agent Layer	1	NOT STARTED	P1	L	58, 45, 40, 37, 48
60	Agents: people, person-name searches, review sites and communities	Agent Layer	1	NOT STARTED	P1	L	58, 46, 41, 48
61	Verification node: Leads, citations and contradictions	Agent Layer	1	NOT STARTED	P0	M	59, 60, 35
62	Distillation and writing-rule engine	Agent Layer	1	NOT STARTED	P0	L	61, 36, 38
63	Model-split experiment per workflow step	Agent Layer	1	NOT STARTED	P2	M	62, 50
64	Document builder: Account Summary and Sources and Evidence	Deliverables & Gate	1	PARTIAL	P1	L	62, 29
65	Workbook builders: People List, Job Postings, Tech Stack and Teams	Deliverables & Gate	1	PARTIAL	P1	M	41, 40, 64
66	Workspace package and Internal-Only file split	Deliverables & Gate	1	PARTIAL	P1	M	64, 65
67	Gate service: pre-delivery checks and loop-back	Deliverables & Gate	1	PARTIAL	P0	L	64, 65, 66, 51
68	Public renderer role and leak detector (Phase A)	Deliverables & Gate	1	PARTIAL	P1	M	23, 64, 67
69	Core API v1	API & Apps	1	PARTIAL	P0	L	28, 56, 31, 29, 67, 57
70	Operator console	API & Apps	1	NOT STARTED	P1	L	69, 57, 32
71	Client portal v1	API & Apps	1	NOT STARTED	P1	L	69, 67
72	Scheduler: cadences, probes and escalation	Scheduler & Alerts	1	NOT STARTED	P1	L	56, 37, 39, 32, 29
73	Alert engine v1: rules, dedupe, digest and email	Scheduler & Alerts	1	NOT STARTED	P1	L	39, 72, 69
74	Observability baseline	Release & Operations	1	NOT STARTED	P1	L	56, 69, 20
75	SLOs, alerting and runbooks	Release & Operations	1	NOT STARTED	P1	M	74
76	Load and chaos test suite	Release & Operations	1	NOT STARTED	P1	L	56, 74, 20, 49
77	Security review and hardening (Phase 1 scope)	Release & Operations	1	NOT STARTED	P1	M	25, 54, 67, 51, 69, 68
78	Parity replay of the nine delivered accounts	Release & Operations	1	NOT STARTED	P0	L	67, 50, 9, 60, 62
79	Production deployment with canary and rollback	Release & Operations	1	NOT STARTED	P1	L	20, 21, 77, 78, 74, 76
80	Technical documentation and runbooks	Docs & Handover	1	NOT STARTED	P1	M	79
81	Client and operator documentation	Docs & Handover	1	NOT STARTED	P1	M	71, 70
82	Concierge pilot launch with two paying tenants	Release & Operations	1	NOT STARTED	P0	L	79, 80, 81, 73, 72, 71, 70, 6, 75
83	Self-serve onboarding and policy-based auto-start	Self-Serve	2	NOT STARTED	P1	L	82, 29, 31, 32
84	Billing and metering with Stripe	Self-Serve	2	NOT STARTED	P1	L	31, 82
85	Custom alert rules and Slack, Teams and webhook delivery	Self-Serve	2	NOT STARTED	P1	L	73, 82

Task	Title	Workstream	Phase	Status	Prio	Effort	Depends on
86	Plain-language question answering over tenant facts	Self-Serve	2	NOT STARTED	P2	L	82, 23, 50
87	Tenant data governance: export, deletion, offboarding and audit viewer	Self-Serve	2	NOT STARTED	P1	L	27, 28, 19, 82
88	Warm-standby region and disaster-recovery game-day	Self-Serve	2	NOT STARTED	P1	XL	82, 17, 18
89	Unit-economics and cost optimisation pass	Self-Serve	2	NOT STARTED	P2	M	82, 13, 74, 63
90	SOC 2 evidence collection	Self-Serve	2	NOT STARTED	P1	M	27, 77, 17, 82
91	Optional shared public cache	Self-Serve	2	NOT STARTED	P3	M	48, 82, 6
92	Per-account vault, tenant keys and bridge and silo tiers	Private Vault	3	NOT STARTED	P1	L	87, 19, 18, 82
93	Internal-source connectors	Private Vault	3	NOT STARTED	P1	XL	92, 5
94	Ingestion pipeline: scan, parse and transcribe	Private Vault	3	NOT STARTED	P1	L	92, 93, 43
95	Redaction policy tiers and redaction maps	Private Vault	3	NOT STARTED	P1	M	94
96	Chunking, embeddings and hybrid retrieval	Private Vault	3	NOT STARTED	P1	L	94, 95, 23, 86
97	Internal fact extraction and relationship state	Private Vault	3	NOT STARTED	P1	M	96, 54, 58, 41
98	Two-output rule: Internal Only view and leakage scan	Private Vault	3	NOT STARTED	P0	L	97, 68, 67
99	Enhanced-version security review and production pilot	Private Vault	3	NOT STARTED	P0	L	98, 90, 88
100	Handover, maintenance plan and 30/60/90-day plan	Docs & Handover	3	NOT STARTED	P2	M	99, 80, 81

Appendix A. Requirements traceability

Requirement	Description	Tasks
FR-01	Self-serve sign-up, organisation, SSO invite	T83, T28, T29
FR-02	Role-based access (admin, seller, viewer)	T28
FR-03	Request a run by name or domain with entity resolution	T69, T58
FR-04	Multi-pass research (baseline, fan-out, distil, verify)	T55, T58, T59, T60, T61, T62
FR-05	Graded brief with citations	T35, T36, T62, T64
FR-06	Blocked, Unverified, Conflicting states shown, never omitted	T35, T47, T38
FR-07	Scheduled refresh with probe then escalate	T72, T37
FR-08	Delta detection and alert rules	T39, T73
FR-09	Alert delivery by email, Slack, signed webhook	T73, T85
FR-10	Credit wallet, top-up, usage, invoices	T31, T84
FR-11	Private vault ingestion	T92, T93, T94, T95, T96
FR-12	Two-output rule	T68, T98
FR-13	Human approval gate	T57
FR-14	Export, deletion, offboarding with crypto-shredding	T87
FR-15	Admin audit-log viewer	T27, T87
FR-16	Operator console	T70
PRD 3.4 Research coverage	Coverage rows with proof, leads need 3 queries across 2 tools	T38, T59, T60, T61, T40
PRD 3.4 People	Deduplicated list, profiles, posts, exclusions	T41, T60, T65
PRD 3.4 Grading and writing	Four grades, plain English, sensitive findings, partner rule	T36, T62
PRD 3.4 Deliverables	Seven-file set with Internal-Only split	T64, T65, T66
PRD 3.4 Quality gate	Pre-delivery checks	T67
PRD 3.5 Standing rules	No outreach, broker names hidden, one account per file set	T67, T66, T32, T62
PRD 4.5 Refresh	Start from previous tracker, reuse source IDs	T72, T39, T48
PRD 3.6 Success metrics	Coverage, traceability, turnaround, spend	T50, T78, T82

Appendix B. Verification register mapped to tasks

ID	Item	Closed by tasks
V1	Model IDs, prices, cache lifetimes	T7, T52, T12, T89
V2	Aurora Global support in ap-south-2, latency	T7, T88
V3	pgvector version, iterative scan, halfvec limits	T7, T10, T96
V4	Inngest limits and SDK signatures	T7, T11, T56
V5	Zero-retention and no-training terms	T6, T94
V6	Speech recognition pricing and diarization	T94
V7	Broker pricing, limits, derived-data use	T6, T13
V8	Karpenter and spot savings	T20, T89
V9	OWASP LLM Top 10 edition	T77
V10	DPDP, GDPR applicability to people data	T6
V11	LangGraph checkpoint setup() and thread deletion	T7, T26, T11

Appendix C. Gate checklist per phase

Phase 0: Decide and Prove

- Isolation spike: zero cross-tenant rows over a 5-connection pool; unset context returns nothing.

- Durability spike: run completes after 5 random kills; each paid call exactly once.
- Prompt cache: read ratio at least 70% from the second call.
- Broker economics: cost per run within cap; coverage at least 0.80.
- ADRs final; conflict register closed; legal memo on file.

Phase 1: Concierge Service

- Isolation suite green on every build; external RLS and injection assessment done.
- Parity replay: at least 95% of High rows match; zero silent omissions.
- Gate service blocks every seeded defect; no broker name in any client-facing file.
- Two paying tenants; one month of scheduled refresh on time; zero isolation defects.
- Canary deploy and rollback proven; runbooks tested by an independent developer.

Phase 2: Self-Serve

- Runs start without an operator within policy; gate failure rate below threshold.
- Billing reconciles to the ledger; no double charge.
- Tenant deletion and offboarding proven; audit viewer live.
- DR game-day within RPO and RTO targets; SLOs met for 30 days; ten paying tenants.

Phase 3: Enhanced (Private Vault)

- Leakage canaries at zero across corpus and 100 builds.
- Retrieval recall and citation targets met per tenant size band.
- DPA signed; vendor retention terms verified; deletion propagates within 24 hours.
- One tenant in production on internal material; SOC 2 Type I issued or scheduled.